INCH-POUND

MIL-T-53101(ME)
7 February 1990

MILITARY SPECIFICATION

TANKS, FABRIC, COLLAPSIBLE: 3,000, 10,000 AND 20,000 GALLON,

PETROLEUM, LOW TEMPERATURE

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tanks, fabric, collapsible: 3,000, 10,000, and 20,000 gallon, petroleum, low temperature, complete with fittings and repair items, for use at all ambient temperatures between -60 °F and +95 °F.

1.2 Classification. Tanks will be 3,000, 10,000 or 20,000 gallon capacity as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
FF-B-561	- Bolt, (Screw), Lag.
FF-B-584	- Bolts, Finned Neck, Key Head, Machine, Ribbed neck, Square Neck, Tee Head.
FF-N-836	- Nut, Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
FF-W-92	- Washer, Flat (Plain).
NN-P-530	- Plywood, Flat Panel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5430

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QQ-P-416	- Plating, Cadmium (Electrodeposited).
RR-C-271	- Chains and Attachments, Welded and Weldless.
TT-W-572	- Wood Preservative: Water-repellent.
VV-F-800	- Fuel Oil, Diesel.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-795	- Cushioning Material Packaging (Flexible Cellular Plastic Film) for Packaging Applications.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-C-1797	- Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam.
PPP-T-60	- Tape, Packaging, Waterproof.
PPP-T-97	- Tape, Packaging/Industrial, Filament Reinforced.

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MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-S-8660	- Silicone Compound, NATO Code Number S-736.
MIL-T-22085	- Tape, Pressure Sensitive, Adhesive, Preservation and Sealing.
MIL-R-25988	- Rubber, Fluorosilicone Elastomer, Oil-and-Fuel-Resistant, Sheets, Strips, Molded Parts, and Extruded Shapes.
MIL-R-25988/1	- Rubber, Fluorosilicone Elastomer, Oil-and-Fuel-Resistant, O-Rings, Class 1, Grade 70.
MIL-B-26195	- Boxes, Wood Cleated, Skidded, Load-Bearing Base.
MIL-R-52255	- Repair Kit and Repair Kit Components for Collapsible Fabric Tanks, Drums, and Boats.
MIL-H-53095	- Hose Assemblies, Rubber, Synthetic, Liquid petroleum Fuels, Dispensing, Collapsible, Low Temperature.
MIL-H-53096	- Hoses and Hose Assemblies, Nonmetallic: Rubber, Liquid Fuel, Noncollapsible, Low Temperature.
MIL-B-53102	- Berm Liner Assembly, Arctic Service.
MIL-T-83133	- Turbine Fuel Aviation, Kerosene, Type Grade JP-8.

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods.
FED-STD-595	- Colors.
FED-STD-601	- Rubber, Sampling and Testing.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-731	- Quality of Wood Members for Containers and Pallets.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing, With Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MS 27019	- Coupling Assembly, Quick Disconnect, Cam-Locking Type.
MS 27022	- Coupling Half, Quick Disconnect, Cam-Locking Type, Male, External Pipe Thread, Type III.
MS 27023	- Coupling Half, Quick Disconnect, Cam-Locking Type, Male, Flanged, Type IV.
MS 27024	- Coupling Half, Quick Disconnect, Cam-Locking Type, Female, Internal Pipe Thread, Type V.
MS 27026	- Coupling Half, Quick Disconnect, Cam-Locking Type, Female, External Pipe Thread, Type VII.
MS 27028	- Coupling Half, Quick Disconnect, Cam-Locking Type, Cap, Dust, Type IX.
MS 27029	- Coupling Half, Quick Disconnect, Cam-Locking Type, Plug, Dust, Type X.
MS 27183	- Washer, Flat-round, Steel, Cadmium Plated, General Purpose.
MS 35338	- Washer, Lock-Spring, Helical.
MS 35649	- Nut, Plain, Hex.
MS 51504	- Elbow, Pipe to tube, 90 Degrees, Male Pipe End, 37 Degrees Flared.
MS 51532	- Cap, Tube, 37 Degrees Flared.
MS 90728	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 8, Cadmium Plated, UNC-2A.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

13228E1768	- Gasket, Quick Disconnect, Arctic Service.
13228E1797	- Valve, Gate.
13228E1827	- Gasket, Flange, Arctic Service.

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(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FSP, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC. (AIA)

NAS 1523 - Packing with Retainer.

(Application for copies should be addressed to the Aerospace Industries Association of America, Inc., 1725 DeSales Street, NW, Washington, DC 20036.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

H35.1 - Alloy and Temper Designation Systems for Aluminum.

(Application for copies should be addressed to the American National Standards Institute, INC., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 26 - Aluminum-Alloy Sand Castings.
B 85 - Aluminum-Alloy Die Castings.
B 108 - Aluminum-Alloy Permanent Mold Castings.
D 381 - Existent Gum in Fuels by Jet Evaporation.
D 412 - Rubber Properties in Tension
D 413 - Rubber Property - Adhesion to Flexible Substrate.
D 429 - Rubber Property - Adhesion to Rigid Substrates.
D 471 - Rubber Property - Effects of Liquids.
D 746 - Brittleness Temperature of Plastics and Elastomers by Impact.
D 750 - Rubber Deterioration in Carbon-Arc or Weathering Apparatus.
D 910 - Aviation Gasolines.
D 1053 - Rubber Property - Stiffening at Low Temperatures: Flexible Polymers and Coated Fabrics.
D 1149 - Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimen).
D 2261 - Fabrics, Woven, By The Tongue (Single Rip) Method (Constant Rate of Extension Tensile Testing Machine), Tearing Strength of.
D 2565 - Operating Xenon Arc-Type (Water-Cooled) Light-Exposure Apparatus With and Without Water for Exposure of Plastics, Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN WOOD PRESERVER'S ASSOCIATION (AWPA)

C1 - All Timber Products - Preservative Treatment by Pressure Process.
C2 - Lumber, Timbers, Bridge Ties and Mine Ties - Preservative Treatment by Pressure Process.
C9 - Plywood-Preservative Treatment by Pressure Process.

(Application for copies should be addressed to the American Wood Preserver's Association, P.O. Box 849, Stevensville, MD 21666.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The assembled collapsible fabric tank shall consist of elastomeric coated fabric tank with attached handles and fittings, repair items, accessories, and a ground cloth.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3. The first article shall include at least one type of repair (of those allowed in 3.13 and in accordance with the repair and rework plan [see 3.13.1]) for validation of each repair. The repair may be applied to either a first article tank or to separate samples.

3.2.1 Sample selection plan. The contractor shall have a sample selection plan prior to initial product. The plan shall be designated for use in the quality conformance testing of section 4 and will provide the Government a method of selecting production seams, bonded fittings, coated fabric, and handle samples for destructive tests. The samples shall be cut from a production tank (see 3.2.2) selected at random in accordance with 4.4.1. The test specimens of seems, bonded fittings, coated fabrics, and handles shall be provided in sufficient quantity for testing and as specified herein. The sample selection plan shall be made available, upon request, to the contracting officer or Government representative on site of manufacture.

3.2.2 Tanks for testing. Tanks selected for samples are in addition to the quantity required for delivery and shall remain the property of the contractor after completion of tests. All tanks selected for the purpose of providing samples for destructive tests are complete with attached handles and fittings. Repair items, accessories, and ground cloth are not included (see 3.2.1).

3.2.3 Additional first article samples. Unless otherwise specified (see 6.2), additional first article samples shall be provided to the U.S. Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-FSH, Fort Belvoir, VA 22060-5606, one week prior to commencement of first article testing. Samples

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shall include all samples required to run tests in tables I, II, III, and IV plus one square yard (in 6 in. x 6 in. x 0.07 to 0.10 in. slabs or equivalent) of each compound used. All samples shall be marked with the applicable contract number.

3.3 Process data. Prior to the initial product, the contractor shall make available to the Government representatives, on site of manufacture, a description of materials, documented work instructions, processing and fabricating procedures which will be used by the contractor to assure that each production tank is a duplicate of the first article tank. Change of materials, components, design, procedures, or sources of supply from those initially selected shall be subject to the approval of the contracting officer (see 6.3).

3.4 Materials. Materials shall be specified herein and as shown on figures 1 through 17. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Tanks shall not be made with materials that fail to meet the requirements of tables I through IV; however, conformance with these requirements shall not be construed as justification for failure to meet other performance requirements of this specification. 1/

3.4.1 Material deterioration prevention and control. The tank shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the tank may exposed.

3.4.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the tanks may be newly fabricated from recovered materials to the maximum extent practicable, provided the tanks produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the tanks.

1/ NOTE: Suggested materials and procedures for tank fabrication may be found in the reports (1) Arctic Fuel Tank Material Development - Final Report, (2) Design and Development of an Arctic 20,000 Gallon Collapsible Fuel Tank (Phase I) and (3) Design and Development of an Arctic 20,000 Gallon Collapsible Fuel Tank (Phase II). These reports are available under Reports No. AD-B113734L, AD-B113823L and AD-B113824L, respectively, from the Defense Technical Information Center, Cameron Station, Alexandria, VA 22304-6145.

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3.4.3 Nylon cloth. The nylon cloth of the coated fabric shall be high-tenacity and heat and light resistant. The nylon cloth shall be free from any imperfection affecting strength, coating adhesion, or coating thickness.

3.4.3.1 Weathering resistance. When tested as specified in 4.5.2.9, the nylon cloth shall have a retained breaking strength that is not less than 45 percent of its initial breaking strength wrap and fill pounds minimum.

3.4.4 Coating compounds. Each and every compound used for the formulation of elastomeric coatings, tie gums, diffusion barrier (if used), and any other materials applied to the coated fabric shall conform to table I and shall be suitable for use with hydrocarbon fuels conforming to MIL-T-5624, MIL-G-3056 (containing up to 40 percent aromatics), ASTM D 910, MIL-T-83133, and VV-F-800. The coating compounds shall be resistant to fungus, weathering, ozone, high humidity, and service in all ambient temperatures between -60 °F and +95 °F.

TABLE I. Characteristics of coating compounds.

Item	Property	Requirement (all tanks)	Test paragraph or test methods of ASTM
1.	Tensile strength (initial) Pull wet within 3 minutes of removal.	1500 lbs/sq. in (min)	D 412
2.	Ultimate elongation	300% (min.)	D 412
3.	Tensile strength after immersion in distilled water at 160 ±2 °F for the following durations:		
	14 days	40% of initial measured tensile strength (min.)	D 471
	42 days	25% of initial	D 471
4.	Tensile strength after fuel immersion in test fluid 1/ at 160 ±2 °F for 14 days.	60% of initial measured tensile strength (min).	D 471
5.	Resistance to light after 1500 hrs accelerated weathering at 10% elongation 2/.	65% of initial tensile strength	D 750 3/ or D 2565 4/
6.	Fuel contamination: 5/ Unwashed existent gum	30 mg/100 mL (max)	4.5.2.11
	Heptane washed existent gum.	7 mg/100 mL (max)	4.5.2.11

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TABLE I. Characteristics of coating compounds. (Cont'd)

Item	Property	Requirement (all tanks)	Test paragraph or test methods of ASTM
7.	Ozone resistance	No cracks under 7X	D 1149 6/
8.	"G" modulus of rigidity after 4 days' exposure at -60 °F.	10,000 lbs/sq. in.	D 1053 7/
9.	Brittleness after 4 days' exposure at -60 °F.	No cracking	D 746

1/ ASTM D 471, Reference Fuel B.
2/ Applicable to all exterior compounds only.
3/ Alternate corex D filters removed.
4/ Method ASTM D 2565, Xenon Arc, procedure A, inner and outer borosilicate filters; deionized water (20 ±3 °C); cycle: 690 minutes light exposure, 30 minutes light and spray; black panel temperature (63 ±3 °C); relative humidity (45 ±5 percent). Specimen rotation every 250 hours.
5/ Applicable to all interior compounds, i.e., compound between the nylon cloth and the inside of the tank.
6/ Test method A specimen shall be conditioned for 7 days at a temperature of 104 ±3.6 °F having a partial pressure of ozone of 50 milipascales (mPa).
7/ Type B specimens.

3.4.5 Coated fabric. The elastomeric coated fabric may include a barrier to restrict diffusion. The coated fabric shall be free from blisters or pinholes and shall show no signs of coating delamination. The coated fabric shall withstand the effects of humidity, fungus, all ambient service temperatures between -60 °F and +95 °F, ozone, and weather elements without damage, deterioration, or failure of meeting performance requirements specified herein. The exterior coating of the coated fabric shall approximate color No. 34094 of FED-STD-595. The elastomeric-coated fabric shall conform to the requirements of table II.

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TABLE II. Characteristics of coated fabric.

Item	Property	Requirement Tank capacity (gallons)			FED-STD-191 or ASTM test method
		3,000	10,000	20,000	
1.	Weight (oz/sq yd)	30 minimum/62 maximum			5041
2.	Diffusion rate (fl oz/ sq ft per 24 hrs [max.])	0.3	0.3	0.3	4.5.2.12
3.	Tearing strength warp and fill (lb [min.])	25	25	35	D 2261
4.	Breaking strength (lbs/in [min.])	350	350	500	5102
5.	Weathering resistance after 2500 hrs exposure at 5% elongation, warp and fill (percent retention of initial breaking strength ([min])	80	80	80	5804/5102 1/ or D 2565 2/
6.	Puncture resistance (lbs [min.])	110	110	150	4.5.2.14/ 5120
7.	Low temperature crease resistance:				
	a. Appearance after unfolding	No cracking, peeling, or delamination			4.5.2.15
	b. Diffusion rate after low temperature crease resistance test 4/ (fl oz/sq ft per 24 hrs [max.])	0.3	0.3	0.3	4.5.2.12
	c. "G" modulus of rigidity after 4 days' exposure at -60 °F (lb/sq. in.)	10,000 (max)			D 1053
8.	Fungus resistance	No cracking, blistering, or delamination of coating. Retention of breaking strength 50% (min.).			5762 3/
9.	Blocking	Specimens to separate within 5 seconds.			4.5.2.16

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TABLE II. Characteristics of coated fabric. (Cont'd)

Item	Property	Requirement Tank capacity (gallons) 3,000	10,000	20,000	FED-STD-191 or ASTM test method
10.	Coating adhesion (initial) (lbs/in [min.]	20	20	20	4.5.2.17 and 4.5.2.17.1
11.	Coating adhesion after immersion in distilled water at 160 ±2 °F for the following duration;				
	14 Days (lbs/in.)	10	10	10	4.5.2.17 and 4.5.2.17.1
	42 days (lbs/in.)	8	8	8	4.5.2.17 and 4.5.2.17.1
12.	Coating adhesion after immersion in test fluid4/ at 160 ±2 °F for 14 days (lbs/in.)	10	10	10	4.5.2.17 and 4.5.2.17.1

1/ Specimens shall have exterior coating facing carbon arc. Alternate corex D filters shall be removed.

2/ Method ASTM D 2565, Xenon Arc, procedure A, inner and outer borosilicate filters; deionized water (20 ±3 °C); cycle: 690 minutes light exposure, 30 minutes light and spray; black panel temperature (63 ±3 °C); relative humidity (45 ±5 percent)

3/ Except that the specimens shall be prepared per method 5102 and the number of specimens shall be reduced from 40 to 5 warp and 5 fill. Leaching of specimens is unnecessary. The specimens shall be exposed to the soil for 8 weeks.

4/ ASTM D 471, reference fuel B.

3.4.6 Gaskets and performed packing (O-ring).

3.4.6.1 Gaskets. Elastomeric gaskets, supplied with quick-disconnect couplings, shall be fluorosilicone gaskets in accordance with 13228E1768. Flange gaskets shown on figures 4 and 5 shall be in accordance with 13228E1827. Gaskets furnished with gate valves (see 3.11) and the gasket furnished with the repair sealing clamp assembly (see 3.9) shall be fluorosilicone gaskets in accordance with MIL-R-25988, type II, class 1.

3.4.6.2 Performed packing (O-ring). Performed packing shown on figures 4, 5, 11 and 14 shall be fluorosilicone O-rings in accordance with MIL-R-25988 and MIL-R-25988/1.

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3.5 Tank construction. The tank shall be fabricated from single-ply coated fabric as specified herein. The configuration and dimensions of the tank and the location of the fittings and handles shall be as shown in figures 1, 2, and 3. For both tank interior and exterior, all coated fabric edges (seems, chafing patches, fabric/flanges of fittings) shall be covered to a thickness of not less than 10 mils with coated fabric or coating compound used to fabricate the tank. Coated fabric panels may be spliced together to make up the length required for a tank panel; however, all such panel splices shall be located on the bottom of the tank. All splices shall conform to all of the requirements of seams as specified in 3.5.1. Splices in adjacent panels shall not coincide.

3.5.1 Seams. All tank seams including end enclosures, handle patches, chafing patches, and fabric flanges of fittings shall conform to the requirements of table III. Seams between adjacent panels of coated fabric shall be constructed to prevent wicking of fuel through the nylon fabric.

TABLE III. Characteristics of seams.

Item	Property	Requirement Tank capacity (gallons)			FED-STD-601 or ASTM test methods
		3,000	10,000	20,000	
1.	Breaking strength (initial) (lbs/in. [min.]) 1/	350	350	500	8311 2/
2.	Breaking strength after immersion in distilled water at 160 ± °F for the following durations:				
	14 days (lbs/in. [min.])	280	280	400	831 1/ 600 1/ 4.5.2.18
	42 days (lbs/in. [min.])	140	140	200	831 1/ 600 1/ 4.5.2.18
3.	Breaking strength after immersion in test fluid 3/ at 160 ±2 °F for the following days (lbs/in. [min.])	315	315	400	831 1/ 600 1/ 4.5.2.18
4.	Dead Load sheer resistance under 50 lb/in. stress at 180 °F for 8 hrs.	0.1 inch slippage (max.)			4.5.2.19

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TABLE III. Characteristics of seams. (Cont'd)

Item	Property	Requirement Tank capacity (gallons) 3,000	10,000	20,000	FED-STD-601 or ASTM test methods
5.	Peel adhesion (initial) (lbs/in. [min.])	20	20	20	D 413 machine method
6.	Peel adhesion after immersion in distilled water at 160 ±2 °F for the following durations:				
	14 days (lbs/in. [min.])	10	10	10	D 413 machine method 6001/4.5.2.18
	42 days (lbs/in. [min.]	8	8	8	D 413 machine method 6001/ 4.5.2.18
7.	Peel adhesion after immersion in test fluid 3/at 160 ±2 °F for 14 days (lbs/in. [min.]	10	10	10	D 413 machine method 6001/ 4.5.2.18

1/ All specimens must break in the coated fabric. Failure of any specimen in a seam area at any value shall constitute failure of this test.
2/ Except that the specimens shall be 2 inches in width.
3/ ASTM D 471, reference fuel B.

3.5.1.1 Seam construction. The seams between adjacent panels of coated fabric shall be constructed to prevent wicking of fuel through the nylon fabric. This shall be accomplished by application of a gum strip inside and outside the tank over the seam so that no edges of the fabric are exposed (except for the inside edges of no more than two final closure seams). The gum strip shall be not less than 0.75 inch wide and not less than 10 mils thick after cure. The gum strip shall be the same compound as the coating to which is bonded. Its bond to the tank and seams, including end and closure splices, shall conform to the requirements of table III when tested as specified in the referenced paragraphs.

3.5.2 Handles. The handles shall be 2-inch wide U-shaped straps fabricated from nylon webbing or cloth and fastened to a coated fabric patch similar to the tank body material. The two ends of each strap shall be attached to each patch

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at points 12 inches, ±1 inch apart. The length of the strap between the two points of attachment shall be 24 inches, ±1 inch. The patch and strap assembly shall be bonded to the bottom of the tank just below the peripheral fold line. Handles shall be located a minimum of 1 inch away from the seams exclusive of end closure seam. The bonds between each handle patch assembly and the tank fabric shall be capable of withstanding perpendicular loads of 1,000 pounds. The number of handles and the position of each handle shall be as shown in figure 1 thru 3, as applicable. Handle assembly color shall approximate color number 34094 of FED-STD-595.

3.5.3 Chafing patches. The interior and exterior of the tank, opposite the location of each fitting, shall be provided with bonded coated fabric chafing patches as shown in figures 1 thru 3. The chafing patches shall be the same coated fabric used to fabricate the tank or shall be of similar coated fabric. Chafing patch material shall not have a deleterious effect on tank material. Chafing patch color shall approximate color number 34094 of FED-STD-595.

3.5.4 Fittings.

3.5.4.1 Fitting type. The access door fitting and the vent and drain flange attachment, as specified herein, shall be bonded to the coated fabric flange and shall conform to the requirements of table IV. Fittings shall be compression type and shall be as shown in figures 6 and 12. The oval closure plate shall be as shown in figures 7 and 8.

TABLE IV. Characteristics of bonded fittings.

Item	Property	Requirement Tank capacity (gallons)			Test Paragraph or ASTM Test Method
		3,000	10,000	20,000	
1.	Aluminum to coated fabric bond strength (initial) (lbs/in. [min.])	350	350	500	4.5.2.20 and 4.5.2.20.1
2.	Bond strength of fitting after immersion in distilled water at 160 ±2 °F for the following durations:				
	14 days (lbs/.in. [min.])	280	280	300	4.5.2.20 and 4.5.2.20.2
	42 days (lbs/in. [min.])	140	140	200	4.5.2.20 and 4.5.2.20.2

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TABLE IV. Characteristics of bonded fittings. (Cont'd)

Item	Property	Requirement Tank capacity (gallons) 3,000	10,000	20,000	Test Paragraph or ASTM Test Method
3.	Bond strength of fitting after fuel immersion in test fluid at 160 ±2 °F for 14 days (lbs/in. [min.])	315	315	400	4.5.2.20 and 4.5.2.20.2
4.	Dead load shear resistance under 50 lbs/in. stress at 180 °F for 8 hours.	0.1 inch slippage (max.)			4.5.2.19 and 4.5.2.20.3
5.	Peel adhesion of aluminum strip to coated fabric (initial) (lbs/in. [min.])	20	20	20	D 429, method B and 4.5.2.21
6.	Peel adhesion of aluminum strip to coated fabric after immersion in distilled water at 160 ±2 °F for the following durations:				
	14 days lbs/in. [min.])	10	10	10	D 429, method B 4.5.2.21
	42 days lbs/in. [min.])	8	8	8	D 429, method B 4.5.2.21, 4.5.2.21.1
7.	Peel adhesion of aluminum to coated fabric after immersion in test fluid 1/ at 160 ±2 °F for 14 days (lbs/in. [min.])	10	10	10	D 429, method B 4.5.2.21, 4.5.2.21.1

1/ ASTM D 471, Reference Fuel B.

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3.5.4.2 Aluminum alloys of tank fittings. Alloy and temper designations of wrought aluminum shall be in accordance with ANSI H35.1. Cast aluminum alloy shall conform to ASTM B 85, UNS A14130 UNS A13600 or UNS A05180 for die castings; ASTM B 108, UNS A03560-T6 for permanent and semipermanent mold castings, or ASTM B 26, UNS A13560T6 or UNS A47120T5 for sand castings.

3.5.4.3 Protective coatings. Aluminum alloy tank fittings shall have protective coatings as specified in figures 7 through 10, 12, and 15. Minimum coating thickness shall be not less than 0.0007 inch for wrought aluminum alloy fittings, and shall be not less than 0.0004 inch for cast aluminum alloy fittings. All metal parts exposed to view on the outside of the tank shall have a dyed coating approximately the tank exterior color and shall be nonvivid and nonreflective.

3.6 Tank performance. The tank and components shall withstand folded storage at ambient temperatures from +145 to -65 °F without damage or leakage when subsequently filled with fuel. The tank and components shall be suitable for operational use at ambient temperatures from +95 to -60 °F. The tank shall not be damaged when exposed to fungus growth or relative humidity up to 100 percent such as is encountered in tropical climates. The tank shall be suitable for use in continuous contact with rainwater, ground water, or water associated with fuels stored in the tank. There shall be no evidence of leakage or seepage when the tank is filled to its rated capacity of diesel fuel, jet fuel, aviation or motor gasoline containing aromatic fuels of up to 40 percent. The tank shall have a 10 percent minimum overcapacity of fuel without rupture or evidence of weakened areas and without leakage or seepage of fuel. The tank shall be capable of withstanding an internal air pressure of 0.50 pound per square inch (psi) without evidence of leakage. The filler assembly elbow shall withstand a hydrostatic pressure not less than 15 psi without leaking.

3.7 Ground cloth. Nylon coated fabric ground cloth shall be furnished with each tank. The ground cloth shall conform to the applicable tank dimensions shown in figures 16 and 17 and shall be fabricated from the single ply coated fabric of the tank (see 3.5). When the ground cloth is to be used with a tank made from polyurethane-coated fabric, the coating must be compatible with the polyurethane and will not contain materials which could react with water to form acid byproducts which would accelerate the deterioration of polyurethane by hydrolysis. The exterior coatings of the coated cloth shall approximate the color specified in 3.4.4.

3.8 Maintainability. The tank shall exhibit no degradation in performance after maintenance operations using the repair parts furnished with the tank.

3.9 Repair Items. The following repair items shall be furnished with each tank (see 3.4.5):

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a. 3,000, 10,000, and 20,000 gallon tanks:

Item	Quantity
Repair kit and repair kit components conforming to MIL-R-52255	1 ea
O-Ring, M25988/1-250, MIL-R-25988/1, MIL-R-25988	2 ea
O-Ring, M25988/1-383, MIL-R-25988/1, MIL-R-25988	2 ea
Gasket, 4-inch flange, 97403/13228E1827-4	2 ea
Gasket, 6-inch flange, 97403/13228E1827-5	2 ea

b. 3,000 gallon tank (in addition to 3.9a):

Item	Quantity
Gasket, quick disconnect coupling 13228E1768-6	2 ea

3.10 Fitting assemblies. Each tank shall be furnished with the following fitting assemblies located as shown in figures 1, 2, and 3. The filler and discharge shall be as shown in figures 4 and 5. The drain fitting assembly shall be as shown in figures 14 and 15. The vent fitting assembly shall be as shown in figure 11. The pressure relief cap of the vent assembly shall be open when subjected to an internal pressure of 3 inches of water. The design of all cam-locking type, quick-disconnect couplings shall conform to MS27019. The maximum torque for all fitting bolts shall be either stamped onto the fitting flange plates or stenciled on the tank fabric adjacent to the respective fittings. Keeper chains as required for the fittings shall conform to RR-C-271, type II, class 3, size 0.035 inch, bronze. Material for the gaskets and preformed packing (O-ring) shall be fluorosilicone (see 3.4.5).

3.11 Accessories.

3.11.1 3,000, 10,000, and 20,000 gallon tank. Unless otherwise specified (see 6.2), each tank shall be provided with the accessories shown in table V.

TABLE V. Accessories provided with each tank.

Item	Conforming to	Quantity 3K, 10K, & 20K
Berm Liner	MIL-B-53102	1
Hose assembly (2 in.)	MIL-H-53096, M53096-06B2A100	2
Hose assembly (4 in.)	MIL-H-53095, M53095-03G200	1
Hose assembly (6 in.)	MIL-H-53096, M53096-10B2A100	1
Gate valve (2 in.)	13228E1797-1	2
Coupling half, male, 2-inch	MS27022-11	2
Coupling half, female, 2-inch	MS27026-11	2
Coupling half, dust cap, 2-inch	MS27028-11	2
Coupling half, dust plug, 2-inch	MS27029-11	2
Gasket, quick disconnect coupling	13228E1768-6	4

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The dust caps and dust plugs shall be attached to the valve assembly. All fittings and valves exposed to view shall approximate the color of the tank coated fabric and shall be nonvivid and nonreflective.

3.12 Identification marking. The tank shall be identified in accordance with MIL-STD-130 by means of an identification label. The label shall be made of coating compound as specified in 3.4.3 or coated fabric as specified in 3.4.4 and shall be bonded to the tank. The following information shall be molded, either recessed or in relief, using 0.50-inch (minimum) letters, on the tank identification label.

TANK, FABRIC, COLLAPSIBLE:
(Specify) GALLONS, PETROLEUM, LOW TEMPERATURE
NSN: (specify)
SERIAL NO.: (Specify)
MFR: (Mfr name and location of plant)
MFG DATE: (Specify month [3 letters] and year)
WEIGHT EMPTY: (Specify approximate number of pounds)
CONTRACT OR ORDER NO: (Specify)
LOT: (Specify)
PART OR IDENTIFYING NUMBER: (See 6.13)

3.12.1 Tank caution label. The tank shall have stencilled or bonded near each inlet a caution label that will withstand weathering. If bonded, the label shall be made of coating compound as specified in 3.4.3 and the color shall approximate No. 34094 of FED-STD-595. The label shall conform to MIL-STD-1472 (labeling) in accordance with 3.15. The following warning shall be in black letters of the height indicated: (Letters shall be 0.75 inch, except as noted below).

CAUTION (1.50-inch letters)

DO NOT OVERFILL (1.50-inch letters)

Overfilling will result in permanent damage and failure of the tank.

Maximum capacity (specify) gallons. Maximum tank height (specify) feet-(specify) inches/(specify) meters.

WARNING: NOT FOR STORAGE OF ALCOHOL DERIVATIVE FUELS (1.50-inch letters)

3.13 Repair and rework of collapsible fabric tanks. Any repair or rework of any tank shall be accomplished before inspection (quality conformance inspection or inspection of packaging, as applicable) with the exception of air leakage testing. Rework or repair of any seam shall not exceed 10 percent of the length of that seam. Repair or rework of panels, other than seams, shall be limited to areas of damage of not greater than six inches in length, six inches in width, or two inches in diameter, as applicable. All repair and rework procedures shall be in accordance with the repair and rework plan.

3.13.1 Repair and rework plan. The contractor shall have a repair and rework plan. The plan shall include a specific description of methods and procedures for repairing and reworking an anomaly which might occur during production of the tank. The plan shall include documented work instructions which shall be used by

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the contractor to ensure that repairs and reworked areas have structural integrity equal to or greater than that of panels and seams, as applicable, that have not been repaired or reworked. No part of the plan shall conflict with or take precedence over the requirements of this specification. The plan shall be made available, upon request, to the contracting officer or Government representative on site of manufacture.

3.14 Workmanship. Workmanship shall be the highest quality and shall permit no defects adversely affecting the strength or serviceability of the finished tank. The fabric-reinforced flange-type fittings shall contain no gum voids, cracks, or tears that could adversely affect the strength of the assembly. All metal parts shall be clean and free of sand, dirt, scale, and flux. Surfaces shall be smooth with edges rounded or beveled. The inside and outside of the tank shall be clean and free of foreign materials. Any necessary repair or rework shall restore the reworked area to its full strength and shall meet all applicable requirements of this specification. The cemented surfaces of all spliced areas, fitting flanges, and patch-type repairs shall effect a bond that will result in strength of the cemented area not less than the strength of adjacent tank fabric.

3.15 Human factors engineering. The tank shall conform to human engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given to General Requirements (4), Controls (5.4), Labeling (5.5), Anthropometry (5.6), Design for Maintainability (5.9), and Hazards and Safety (5.13) of MIL-STD-1472, as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Process inspection. Inspection shall be accomplished by the contractor to assure compliance with the approved documented work instructions covering processing and fabricating procedures. Any process or fabricating procedure not

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in accordance with the contractor's procedure used in manufacture shall be cause to withhold acceptance of tanks determined to have been manufactured using the unapproved process or fabricating procedure.

4.2 Classification of inspection. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6)
d. Inspection of packaging (see 4.7).

4.3 First article inspection. Unless otherwise specified, the first article inspection will be performed by the contractor. Acceptance of the first article tanks shall not exclude the remaining tanks from meeting all the requirements of this specification. In addition to the performance of first article inspections, the Government reserves the right to perform any and all inspections contained in this specification to assure compliance. Failure to meet any requirement of this specification shall constitute failure of first article inspection.

4.3.1 Examination. The first article tank shall be examined as specified in table VI. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article tank shall be subjected to tests marked "X" in column 1 of table VII. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for destructive test. Selection of samples for destructive tests shall be in accordance with the approved sample selection plan (see 3.2.1). The frequency for random selection of production tanks will be once for each 26 tanks produced. The 26 tanks will be identified as a lot corresponding to the selected samples. The tank used for destructive testing shall remain the property of the contractor. The remaining 25 tanks are deliverable tanks in that lot. Continuous successful completion of four test cycles (see 4.4.2) may be cause for reducing the frequency of testing, subject to approval of the contracting officer. Any reduction in frequency will cause a corresponding increase in the represented lot size. Rejection of any lot (see 4.4.2) during the reduced selection frequency will require a return to the original frequency (see 3.2.1).

4.4.2 Destructive tests. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table VII. Bonded fittings and seams test samples shall be cut from a completed production tank. Failure of any test shall be cause for rejection of represented lot of tanks.

4.4.3 Complete tank. Each complete tank assembly shall be examined as specified in table VI and subjected to the tests marked "X" in column 2 of table VII. Any nonconformance revealed by the examination or failure of the test shall be cause for rejection of the tank.

4.5 Inspection procedure.

4.5.1 Examination. The tank shall be examined for the following:

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TABLE VI. Examination schedule.

Examination No.	Examination description	Requirement Paragraph
101.	Sample selection plan not available.	3.2.1
102	Rework and repair plan not available.	3.1.3.1
103	Process data not available.	3.3
104	Changed process data not approved by the contracting officer.	3.3
105.	Materials not as specified.	3.4
106.	Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.4.1
107.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each.	3.4.1.1
108.	Contractor does not have documentation available for identification of material, material finishes or treatment.	3.4.1.2
109.	Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the tanks.	3.4.2
110.	Barrier film loose or evidence of coating delamination.	3.4.5
111.	Blisters or pinholes in coated fabric.	3.4.5
112.	Exterior color of tank not as specified.	3.4.5
113.	Gaskets and performed packing not as specified.	3.4.6
114.	Dimensions not as specified.	3.5
115.	Edges of coated fabric not covered as specified.	3.5
116.	Splices and seams coincident or seams coincident with tank fittings.	3.5 & 3.5.1
117.	Handles, chafing patches, or fittings located as specified.	3.5, 3.5.2, 3.5.3 & 3.5.4
118.	Fittings not as specified.	3.5.4 , 3.10
119.	Finish of fittings not as specified.	3.5.4.3
120.	Ground cloth not as specified.	3.7
121.	Repair items missing or not as specified.	3.9
122.	Keeper chains not as specified.	3.10
123.	Maximum torque for bolts not listed on tank.	3.10
124.	Accessories not as specified.	3.11
125.	Identification label not as specified.	3.12
126.	O-rings not lubricated as specified.	Figures 4, 5, 11 & 14
127.	Identification marking missing, incomplete, or illegible.	3.12

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TABLE VI. Examination schedule. (Cont'd)

Examination No.	Examination description	Requirement Paragraph
128.	Tank caution label missing or not as specified.	3.12.1
129.	Extraneous material inside tank.	3.14
130.	Panel splices not as specified.	3.5
131.	Workmanship not as specified.	3.14
132.	Human factors engineering not as specified.	3.15
133.	Rework or repair, as applicable, not as specified in rework and repair plan.	3.13.1
134.	Repair or rework of seams or tank body not as specified.	3.13.1

4.5.2 Tests. The tank shall be tested for the following:

TABLE VII. Test schedule.

First article	Quality conformance: Individual	Quality conformance: Sample (see 3.2.1)	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
			Tank and accessories 1/		
X	-	X	Vent fitting assembly.	4.5.2.1	3.10
X	X	-	Air leakage.	4.5.2.2	3.6
X	-	-	Hydrostatic.	4.5.2.3	3.6
X	-	-	Low temperature.	4.5.2.4	3.6
X	-	-	High temperature.	4.5.2.5	3.6
X	-	-	Fuel storage.	4.5.2.6	3.6
X	-	-	Overload.	4.5.2.7	3.6
X	-	-	Internal inspection.	4.5.2.7.1	3.6
X	-	X	Handle pull test.	4.5.2.8	3.5.2
			Nylon cloth		
X	-	-	Weathering resistance.	4.5.2.9	3.4.3.1
			Coating compounds		
X	-	X	Tensile strength.	4.5.2.10	3.4.4, table I
X	-	X	Ultimate elongation.	4.5.2.10	3.4.4, table I
X	-	X	Tensile strength after 14-day water immersion.	4.5.2.10	3.4.4, table I
X	-	-	42-day water immersion.		

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TABLE VII. Test schedule. (continued)

First article	Quality conformance: Individual	Quality conformance: Sample (see 3.2.1)	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	-	X	Tensile strength after 14-day fuel immersion.	4.5.2.10	3.4.4, table I
X	-	-	Resistance to light.	4.5.2.10	3.4.4, table I
X	-	-	Unwashed existent gum.	4.5.2.11	3.4.4, table I
X	-	-	Heptane washed gum.	4.5.2.11	3.4.4, table I
X	-	-	Resistance to ozone.	4.5.2.10	3.4.4, table I
X	-	-	"G" modulus of rigidity.	4.5.2.10	3.4.4, table I
X	-	-	Brittleness.	4.5.2.10	3.4.4, table I
			Coated fabric		
X	-	X	Diffusion rate.	4.5.2.12	3.4.5, table II
X	-	X	Weight.	4.5.2.13	3.4.5, table II
X	-	X	Tearing strength.	4.5.2.13	3.4.5, table II
X	-	X	Breaking strength.	4.5.2.13	3.4.5, table II
X	-	-	Weathering resistance.	4.5.2.13	3.4.5, table II
X	-	-	Fungus resistance.	4.5.2.13	3.4.5, table II
X	-	X	Puncture resistance.	4.5.2.14	3.4.5, table II
X	-	X	Low temperature crease resistance.	4.5.2.15	3.4.5, table II
X	-	X	Blocking.	4.5.2.16	3.4.5, table II
X	-	X	Coating adhesion (initial).	4.5.2.17.1	3.4.5, table II
X	-	X	Coating adhesion after 14-day water immersion	4.5.2.17.1	3.4.5, table II
X	-	-	Coating adhesion after 41-day water immersion.	4.5.2.17.1	3.4.5, table II
X	-	X	Coating adhesion after 14-day fuel immersion.	4.5.2.17.1	3.4.5, table II

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TABLE VII. Test schedule. (continued)

First article	Quality conformance: Individual	Quality conformance: Sample (see 3.2.1)	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
			Seams		
X	-	X	Breaking strength (initial).	4.5.2.18	3.5.1, table III
X	-	X	Breaking strength after 14-day water immersion.	4.5.2.18	3.5.1, table III
X	-	-	Breaking strength after 42-day water immersion.	4.5.2.18	3.5.1, table III
X	-	X	Breaking strength after 14-day fuel immersion.	4.5.2.18	3.5.1, table III
X	-	X	Dead load shear resistance.	4.5.2.19	3.5.1, table III
X	-	X	Peel adhesion (initial).	4.5.2.18	3.5.1, table III
X	-	X	Peel adhesion after 14-day water immersion.	4.5.2.18	3.5.1, table III
X	-	-	Peel adhesion after 42-day water immersion.	4.5.2.18	3.5.1, table III
X	-	X	Peel adhesion after 14-day fuel immersion.	4.5.2.18	3.5.1, table III
			Bonded fittings		
X	-	X	Bond strength (initial).	4.5.2.20 and 4.5.2.20.1	3.5.4.1, table IV
X	-	X	Bond strength after 14-day water immersion.	4.5.2.20 and 4.5.2.20.2	3.5.4.1, table IV
X	-	-	Bond strength after 42-day water immersion.	4.5.2.20 and 4.5.2.20.2	3.5.4.1, table IV
X	-	X	Bond strength after 14-day fuel immersion.	4.5.2.20 and 4.5.2.20.2	3.5.4.1, table IV
X	-	X	Dead load shear resistance.	4.5.2.20 and 4.5.2.20.3	3.5.4.1, table IV
X	-	X	Peel adhesion of aluminum strip (initial).	4.5.2.21 and 4.5.2.21.1	3.5.4.1, table IV
X	-	X	Peel adhesion after 14-day water immersion.	4.5.2.21 and 4.5.2.21.1	3.5.4.1, table IV
X	-	-	Peel adhesion after 42-day water immersion.	4.5.2.21 and 4.5.2.21.1	3.5.4.1, table IV
X	-	X	Peel adhesion after 14-day fuel immersion.	4.5.2.21 and 4.5.2.21.1	3.5.4.1, table IV

1/ The tank assembly shall be tested in the order listed starting with the air leakage test.

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4.5.2.1 Vent assembly. Subject to vent assembly to an internal pressure of 3 inches of water. Inability of the pressure relief cap to open at this pressure shall constitute failure of this test.

4.5.2.2 Air leakage. Pressurize each tank to a 0.50 psi, ±5 percent, internal air pressure and allow to stand for 0.50 hours. At the end of this period, adjust the internal air pressure to 0.50 psi, ±5 percent. Then, using a soap and water solution, examine all the tank fabric, fittings, and external seams for leakage. Any evidence of air leakage shall constitute failure of this test.

4.5.2.3 Hydrostatic. Subject the elbow filler assembly to a hydrostatic pressure of 15 psi for a period of not less than 10 seconds. The test fluid shall be water. Any evidence of leakage of the assembly shall constitute failure of this test.

4.5.2.4 Low temperature. The tank shall be folded or rolled to a size suitable for placing in the shipping container as specified in 5.3 and 6.2 and then placed in a low temperature storage environment of -65 ±2 °F for a period of 24 hours. The environmental temperature shall then be increased to -60 °F for an additional 24 hours. At the end of this period and while still at -60 °F, the tank shall be unfolded in not less than 15 minutes not more than 30 minutes. Any flaking, cracking, or separation of the coated fabric shall constitute failure of this test.

4.5.2.5 High temperature. The tank shall be folded or rolled to a size suitable for placing in the shipping container as specified in 5.3 and 6.2 and then placed in a high temperature storage environment of 145 ±2 °F for a period of 24 hours. The higher environmental temperature shall then be lowered to +95 °F for an additional 24 hours. At the end of this time and while still at +95 °F, the tank shall be unfolded in not less than 15 minutes nor more than 30 minutes. Any flaking, cracking, delamination, or separation of the coated fabric shall constitute failure of this test.

4.5.2.6 Fuel storage. Fill the tank outdoors without any environmental protective covering with its rated gallon capacity of gasoline (MIL-G-3056), jet fuel (MIL-T-5624 or MIL-T-83133), diesel fuel (VV-F-800), or aviation gas (ASTM D 910) as specified (see 6.2) and allow to stand for 30 days, ±2 days. At the end of this period, examine the tank for seepage and leakage. When emptied, the tank shall be rolled up and tank bottom examined for leakage or seepage. Any evidence of seepage shall constitute failure of this test.

4.5.2.7 Tank overload. In preparation for the overload tests, the tank shall be emptied of the fuel and allowed to lie empty outdoors for 5 days, ±1 day, without any environmental protective covering. The tank shall be located on a non-absorbent ground cloth during this interval. The tank shall then be filled to its rated capacity +10 percent of gasoline (MIL-G-3056); jet fuel (MIL-T-5624 or MIL-T-83133), diesel fuel (VV-F-800), or aviation gas (ASTM D 910) as specified (see 6.2) and allowed to stand for 4 hours in a dry location. When emptied, the tank shall be rolled up and tank bottom examined for leakage or seepage. Any evidence of rupture, weakened areas, leakage, or seepage shall constitute failure of this test.

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4.5.2.7.1 Internal inspection. The tank shall be inspected internally after the overload test of 4.5.2.7. Any evidence of weakened areas, coating, or barrier delamination shall constitute failure of the overload test.

4.5.2.8 Tank handle pull resistance. The test sample shall consist of the handle, coated fabric patch, and a minimum of 1 foot tank body fabric extending in all directions from the handle patch. The body fabric shall be tightly drawn and clamped between two flat oval rings or a base plate and an oval ring so that the handle of the patch is centrally located in the rings. The shape of the oval rings shall be such that all edges of the coated fabric patch are 6 inches, ±1 inch, from the ring clamp. The rigidly, strength, and construction of the clamp shall be such that the tank body material shall not slip more than 0.50 inch at any point during the test. With the sample held securely, a tension shall be applied through a bar or pipe 1 inch in diameter inserted in the loop of the handle. The tension shall be applied slowly and smoothly in a direction perpendicular to the plane of the handle patch until the specified load of 1,000 pounds is reached. The 1,000 pound load shall be maintained for 1 minute. Any damage, permanent distortion, or separation of the handle patch or tank material shall constitute failure of this test.

4.5.2.9 Nylon cloth. Nylon cloth properties shall be tested in accordance with FED-STD-191, method 5804 and 5104. Properties shall apply after heat setting. In method 5804 alternate corex D filter shall be removed. Specimens shall be raveled for method 5104 after 100 hours if accelerated weathering. The initial breaking strength warp and fill (lbs [min.]) shall be determined. The edges of the test specimen shall be coated by dipping or brushing with a nylon solution or other adhesive or sealed by melting with heat sufficiently to preclude yarn slipping while under test (see 6.5). Breaking strength after accelerated weathering less than 45 percent of the initial breaking strength shall constitute failure of this test.

4.5.2.10 Coating compounds. Coating compound properties shall be tested in accordance with test methods shown in table I. Immersed specimens shall be conditioned at room temperature for 30-90 minutes in fresh immersion media prior to testing. Nonconformance to 3.4.3 or table I shall constitute failure of this test.

4.5.2.11 Existent gum. The test fluid shall be prepared according to ASTM D 471, Reference Fuel B. Cut a 5-gram specimen of each tank interior coating compound into approximately 0.0625-inch squares and place in a flask containing 250 mL of test fuel and allow to stand for 48 hours at 73 ±3 °F (23 ±2 °C). Decant and filter the contaminated fluid through Whatman 41H filter paper or equal. Determine the unwashed existent gum and the heptane washed gum content of the filtrate in accordance with ASTM D 381, using the air jet vaporizing medium and an evaporation time of 45 minutes. A sample of unused test fuel will be tested as a blank and subtracted from the samples containing tank material. A minimum of three specimens shall be used to report the average gum content. Nonconformance to 3.4.4 and table I shall constitute failure of this test.

4.5.2.12 Diffusion. The test apparatus shall consist of a diffusion cup and ring, using figure 13 as a guide. Other cup designs are acceptable as long as the inside diameter of the cup and ring is maintained at 2.00 inches, ±.016 inch, and the cup is approximately 1.00 inch deep (minimum). Cut a circular test disk of coated fabric to conform to the outside diameter of the cup flange. Punch

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holes in the disk to correspond to the flange bolt dimensions as needed. The cup shall be filled with approximately 40 mL of Reference Fuel B in accordance with ASTM D 471. A suitable solution (see 6.5) shall be used to seal the exposed fabric around the outer edge of the test disk and also for sealing the test disk to the diffusion cup flange. The test disk shall be placed over the cup with the tank "interior" side toward the fuel. The bolts shall be tightened securely. Place the diffusion cup in a suitable rack in a constant temperature of 73.5 ±2 °F and a relative humidity of 65 percent. Allow 1 hour for the assembly to reach equilibrium, then weight the cup to the nearest 0.005 gram and place in the rack face upward. Keep the cup at the above constant temperature and humidity for 24 hours. Then weigh the check for vapor loss. Retorque the bolts if necessary. Invert the cup (test disk down) in a rack that permits free access of air to the test disk. Weigh the cup daily. Defective films or leaks caused by faulty assembly are usually found when the cup is weighed on the third day. Continue to weigh the cup daily until the weight loss is constant to within 0.010 gram per day after two 24-hour periods. Then record daily weight loss for a continuous interval of 72 hours. The diffusion rate (D) in fluid ounces per square foot per 24 hours shall be the average of not less than three specimens when calculated from the following expression:

$$D = \frac{144 \text{ (Average daily loss in grams)}}{\text{(Sp. gr.) (29.573) (3.142) } (R^2)}$$

Where Sp. gr. is the specific gravity of the test medium and R is the inside radius of the test cup in inches.

4.5.2.12.1 Failure criteria. Nonconformance to 3.4.4 and table II shall constitute failure of this test.

4.5.2.13 Coated fabric. Coated fabric properties be tested in accordance with test methods shown in table II. Nonconformance to 3.4.4 and table II shall constitute failure of this test.

4.5.2.14 Puncture resistance. FED-STD-191, method 5120, applies except that the ring clamp mechanism shall have an internal diameter of 3.00 inches, and the ball shall be replaced by a piercing instrument shaped like a flared, flat-tip screwdriver, having a width of 0.312, ±0.312, ±0.010 inch, and a thickness of 0.031, ±0.004 inch, at the extreme tip. The piercing tip edges shall be rounded to a 0.010 inch radius. The piercing instrument shall be oriented to intercept the warp and fill threads at an angle of approximately 45 degrees. The average of three test specimens shall be reported. Nonconformance to 3.4.4 and table II shall constitute failure of this test.

4.5.2.15 Low temperature crease resistance. Prepare test fluid according to ASTM D 471, reference fuel B. Immerse three coated fabric specimens, each 8 inches square, in the test fluid and condition for 24 hours, ±1 hour, at 73 ±5 °F. Remove specimens, place on a wire screen, and air dry with forced air movement at 73 ±5 °F, for 24 hours, ±1 hour. Then fold the specimens in half in each direction so that a folded corner occurs in the center of each specimen. Place each folded specimen under a 4-pound load and condition at -60 °F for 46 hours. At the end of the conditioning period, unfold the specimens while still at a temperature of -60 °F and examine visually. Signs of cracking, peeling, or delamination of any coating material shall constitute failure of this test. If the specimen does not fail, then subject the specimen to the diffusion test specified in 4.5.2.12 except position the specimen in the diffusion cup in such a

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manner that the center of the previously folded specimen coincides with the center of the cup. Nonconformance to 3.4.4 and table II shall constitute failure of this test.

4.5.2.16 Blocking. Place three coated fabric specimens 6 inches by 1 inch in an oven on a smooth surface in such a manner that the ends are overlapped 1 inch. Place a 4-pound weight directly on the overlapped area. After conditioning at a temperature of 158 ±2 °F for 4 hours, remove the weight and take the specimens from the oven and condition for 1 hour at 73 ±3 °F. Attach one end of the specimen in a suitable clamping device allowing the free end to hang down. Suspend a 4-ounce weight from the free end of the specimens. Inability of the strips to separate within 5 seconds under the 4-ounce load shall constitute failure of this test.

4.5.2.17 Coating adhesion. Samples of coated fabric shall be bonded face-to-face to provide specimens for determining adhesion between the nylon cloth and exterior coatings (s), between the nylon cloth and interior coatings(s), between laminations of interior coatings and barrier (if used), and between laminations of exterior coatings. In forming this bond, the specimens shall be subjected to no heat or pressure other than that normally encountered in curing the coated fabric, except for minimal pressure necessary to insure contact while the bond is setting.

4.5.2.17.1 Test procedure. The adhesion shall be determined in accordance with ASTM D 413, machine method, except that the specimens shall be 2 inches wide. The specimens shall be of sufficient length to conduct adhesion test for both initial values and after fuel or water immersions. The adhesion results obtained on each immersed specimen shall be compared with the initial adhesion of the same specimen to determine percentage of adhesion retained. The reported adhesion and percent retention shall be the average of not less than three specimens. Attempts shall be made to cut the coating back to the nylon cloth and to determine the adhesion value at the coating-to-cloth interface. However, if a specimen separates at a plane other than the bond of the coating-to-cloth (such as between layers of coating materials or between barrier film and coating). The adhesion value and the plane of failure shall be recorded. Immersed specimens shall be conditioned in the test fluid at 73 ±5 °F for 30 to 90 minutes before testing. Testing of immersed specimens shall be completed within 3 minutes after removal from the conditioning fluid. Immersion of specimens shall be in accordance with ASTM D 471. Nonconformance to 3.4.4 and table II shall constitute failure of this test. Any obvious bond failure evident after immersion but before stressing, even if the plane of failure is not sandwiched between the layers of nylon fabric, shall constitute failure of this test.

4.5.2.18 Seam tests. The bonding together of any two or more pieces of coated fabric (such as lap joints, butt joints, and closures, chafing patches, coated fabric flanges of fittings, etc.) shall be considered as seams and shall be subjected to all seam tests specified herein, except chafing patches which will only be tested for peel. The average breaking strength of five specimens for each type seam for each test shall be reported for conformance to table III. Breaking strength specimens shall be 2 inches wide (parallel to the seam) and shall extend (perpendicular to the seam) 3 inches beyond both edges of the seam. No part of the test specimens shall be coated or covered during the fuel or water immersion periods. Specimens shall be cooled in the immersion fluids at 73 ±5 °F for 30 to 90 minutes before testing. Testing of immersed specimens shall be

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completed within 3 minutes after removal from the immersion fluids. The average peel adhesion strength of three specimens for each type seam shall be reported for conformance to table III. Peel adhesion specimens shall be of sufficient length to determine both the initial and after fuel or water adhesion values on the same specimen. Nonconformance to 3.5.1 and table III shall constitute failure of this test.

4.5.2.19 Dead load shear resistance. The test specimens shall be 1.0 inch, ±0.020 inch, wide (parallel to the same) and coated fabric shall extend a minimum of 3 inches (perpendicular to the seam) on each side of the seam. One index mark shall be scribed on each side of the seam to facilitate observation and measurement of slippage. Each specimen shall be subjected to a constant (dead load) tension force of 50 ±0.50 pound at 180 ±5 °F. After 8 hours, examine each specimen while still under tension for sign of slippage or separation. Three specimens shall be tested for each determination. Slippage, by any specimen, greater than specified in 3.5.1 and table III shall constitute failure of this test.

4.5.2.20 Strength of bonded fittings. Specimens shall be prepared by cutting through the aluminum flange so that parallel 1.0-inch wide sections are obtained from the straight portion of the oval fitting and 1.0-inch wedge shaped sections are obtained from the vent (or drain) and the curved portion of the oval fitting. The 1.0 inch shall be measured as a chord passing through the midpoint between the inside and outside diameters of the flange for the wedge shaped sections. If contractor's alternate fittings are specified, samples shall be cut similarly to the above description providing 1.0-inch specimens measured at a chord midway between the internal and external radii.

4.5.2.20.1 Initial bond strength. The coated fabric flanges shall be fastened together in one jaw of the test machine so that the jaw will be at least 1.0 inch from the nearest part of the aluminum flange. The aluminum flange shall be secured in the other jaw of the test machine and this jaw shall clamp on the aluminum and shall not compress the embedded part of the coated fabric flanges. The jaws shall be separated at a rate of 2.0 inch per minute at 75 ±5 °F. the average of three test specimens shall be recorded as initial bond strength in pounds per inch of width. Nonconformance to 3.5.4.1 and table IV shall constitute failure of this test.

4.5.2.20.2 Bond strength after fluid immersion. Three test specimens shall be immersed for the appropriate durations in each test fluid specified in table IV. No part of the specimens shall be covered or coated during immersion. Specimens from both the oval and vent/drain fittings shall be included in each test fluid. The test specimens shall be cooled in the immersion fluid to 75 ±5 °F for 30-90 minutes. The specimens shall be removed from the test fluid, one at a time, and tested as in 4.5.2.20.1. Each test shall be completed within 3 minutes after removal from the test fluid. The average of three tests for each fluid shall be reported as bond strength after immersion in pounds per inch of width. Nonconformance to 3.5.4.1 and table IV shall constitute failure of this test.

4.5.2.20.3 Dead load shear resistance aluminum-to-fabric bond. Three specimens shall be clamped as in 4.5.2.19 and subjected to a constant (dead load) tension force of 50 pounds at 180 ±5 °F. At the end of 8 hours, the specimens shall be examined for slippage or separation while under tension. Nonconformance to 3.5.4.1 and table IV shall constitute failure of this test.

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4.5.2.21 Peel adhesion of aluminum-to-coated fabric. Special test specimens shall be fabricated consisting of aluminum strips bonded to lengths of coated fabric. The aluminum strip shall be 12 inches long and shall be of the same alloy as that used in the aluminum fitting flanges. The coated fabric shall be 12 inches long (minimum) by 2.0 inches, +0.050 inch, wide and shall be of the same composition (and of the same state of cure before bonding) as that used in the coated fabric flanges. The coated fabric strip shall be uniformly bonded to the aluminum strip. The bond shall be formed using identical techniques and bonding agents used to bond tank fittings and shall be cured identically (time, pressure, temperature, etc.) to the process used in bonding tank fittings.

4.5.2.21.1 Test procedures. Specimens shall be tested as per ASTM D 429, method B. Three specimens shall be averaged for each initial and fluid immersion test. The same identical specimens shall be used to determine the initial peel strength, the strength after fluid immersion, and when computing the percentage of initial adhesion retained. Immersed specimens shall be conditioned for 30-90 minutes at room temperature in fresh immersion medium prior to testing. Nonconformance to 3.5.4.1 and table IV shall constitute failure of this test.

4.6 Inspection comparison. The Government may select tanks at any time during the contract production period and subject these tanks to the examinations and tests specified herein. The inspection will be performed by the Government at a site selected by the Government. Tanks for inspection will be selected at random from previously selected product but will not include first article tanks.

4.6.1 Inspection failure. Failure of an inspection comparison tank to meet any requirements specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison tank and shall be cause for refusal by the Government to continue acceptance of production tanks until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on tanks previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all tanks accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 First article packaging inspection. The first article pack shall be inspected as follows:

4.7.1.1 Examination. The first article pack shall be examined for the defects listed in 4.7.2.3. The presence of one or more defects shall be cause for rejection.

4.7.1.2 Tests. The first article pack shall be tested as follows:

4.7.1.2.1 Rail impact. When level A packing is required (see 6.2), the first article packs shall be subjected to the rail impact test specified in MIL-STD-1186 (see 5.4). During the unpacking test (see 4.7.1.2.2), the blocking

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and bracing method shall be evaluated. Any longitudinal movement of the contents causing damage to the contents or the shipping container which will render the contents or the container unless for its intended use, shall constitute failure of the test.

4.7.1.2.2 Unpacking. The first article pack, for all levels of packing, shall be unpacked in the manner and sequence prescribed by the contractor. The first article pack shall have failed the test should the prescribed manner and sequence be ineffective, or should any of the contents fall free of the package.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

No.	Defect
135.	Repair Kit not preserved as specifiedfor level A (see 5.2.1.1).
136.	Other emergency repair items not preserved as specified for level A (see 5.2.1.2).
137.	Preserved emergency repair items not combined together in preconsolidation box as specified for level A (see 5.2.1.3).
138.	Preconsolidation box not as specified for level A (see 5.2.1.3).
139.	Contents of preconsolidation box not cushioned as specified for level A (see 5.2.1.3).
140.	Marking of preconsolidation box not as specified for level A (see 5.2.1.3).
141.	Fittings not preserved as specified for level A (see 5.2.2.
142.	Hose assemblies not preserved as specified for level A (see 5.2.3.1).
143.	Other accessories not preserved as specified level A (see 5.2.3.2).
144.	Technical publication, when required, not preserved as specified for level A (see 5.2.4).
145.	Consolidation of all components not as specified for level A (see 5.2.5).
146.	Consolidation box not as specified for level A (see 5.2.5).
147.	Consolidation box not marked as specified for level A (see 5.2.5).
148.	Cushioning of consolidation box contents not as specified for level A (see 5.2.5).
149.	Openings into tank not protected as specified for level A (see 5.2.6).
150.	Tank not dusted with talc or other anti-sticking compound as specified for level A (see 5.2.6).
151.	Permanently attached tank fittings not preserved as specified for level A (see 5.2.6).
152.	Tank not folded or rolled into a compact bundle as specified for level A (see 5.2.6).
153.	Bundle not secured as specified for level A (see 5.2.6).

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154. Ground cloth, not preserved as specified for level A (see 5.2.7).
155. Tank of urethane-coated fabric not preserved as specified for level A (see 5.2.8).
156. Ground cloth of urethane-coated fabric not preserved as specified for level A (see 5.2.8).
157. Lumber and plywood not treated with preservative as specified for level A (see 5.3.1.1).
158. Carriage bolts, lag screws, nuts and washers not as specified for levels A (see 5.3.1.2 and 5.3.2).
159. Plywood not as specified for levels A and B (see 5.3.1.2.1 and 5.3.2).
160. Cleats not as specified for levels A and B (see 5.3.1.2.2 and 5.3.2).
161. Skids and rubbing strips not as specified for levels A and B (see 5.3.1.2.3 and 5.3.2).
162. Fabrication of shipping container specified for levels A and B (see 5.3.1.2.4 and 5.3.2).
163. Anti-abrasion provisions not as specified for levels A, B, and C (see 5.3.1.2.5, 5.3.2, and 5.3.3).
164. Consolidation containers not positioned and secured as specified for levels A, B, and C (see 5.3.1.2.6, 5.3.2, and 5.3.3).
165. Wire-reinforced hose assemblies not positioned and secured as specified for levels A, B, and C (see 5.3.1.2.7, 5.3.2, and 5.3.3).
166. Containers with outside length dimension 96 inches or less not as specified for levels A and B (see 5.3.1.2.8 and 5.3.2).
167. Containers with outside length greater than 96 inches not as specified for levels A and B (see 5.3.1.2.9 and 5.3.2).
168. Shipping container for level C packing not as specified for level C (see 5.3.3).
169. Contents of shipping container not blocked and braced as specified for levels A, B, and C (see 5.4).
170. Marking not as specified for levels A, B, and C (see 5.5).

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination and test within the time frame specified (see 6.2) to prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3.1). The first article pack may be accomplished utilizing either the first produced collapsible tank or a production collapsible tank. If the first produced collapsible tank is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first produced collapsible tank and the production collapsible tank.

5.2 Preservation. Each complete collapsible tank shall be preserved to satisfy level A preservation requirements in the following manner:

5.2.1 Emergency repair items.

5.2.1.1 Repair kit. Each component of the repair kit shall be preserved in accordance with the level A requirements of MIL-R-52255.

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5.2.1.2 Other emergency repair items. The remaining emergency repair items shall be preserved in accordance with MIL-P-116, method IC-1, or IC-3 and items of like description for each tank shall be preserved together, in the quantities specified, in such a manner as to prevent damage or distortion.

5.2.1.3 Preconsolidation. The preserved emergency repair items, including the preserved repair kit components, shall be consolidated together in a close-fitting fiberboard box conforming to PPP-B-636, class weather resistant, grade as applicable. Cushioning shall be positioned within the box to prevent free movement and damage to the contents. Box closure shall be in accordance with method IV of the appendix to the box specification. The box shall be legibly marked as follows:

EMERGENCY REPAIR ITEMS
FOR
COLLAPSIBLE PETROLEUM TANKS

5.2.2 Fittings. Removable portions of all quick-disconnect type fittings shall be separated from the tank at a quick-disconnect point. Each separated portion shall be wrapped with cushioning material conforming to PPP-C-795 (with the exception that the Creep test specified in PPP-C-795 may be eliminated), PPP-C-1797 or PPP-C-843, type II, class B, water-resistant. The cushioning material shall be secured in place with tape conforming to MIL-T-22085, type II, PPP-T-60, type III or IV, or PPP-T-97, type IV, and the wrapped fittings shall be marked for identification.

5.2.3 Accessories. When provided (see 6.2), the accessories shall be preserved as follows:

5.2.3.1 Hose assemblies. Each hose, assembled as specified and with the cap and plug positioned as specified, shall have the metal portions of all fittings securely wrapped with cushioning material as specified in 5.2.2. Nonwire-reinforced hose assemblies shall be coiled to the smallest coil diameter without causing a permanent set or damage to the hose and secured as a coil with twine at three points equidistant around the circumference of the coil.

5.2.3.2 Other accessories. The remaining accessories shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3 and accessories of like description for each tank shall be preserved together in the quantities specified.

5.2.4 Technical publications. Technical publications, when required, shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.5 Consolidation. The preserved emergency repair items, preconsolidated as specified in 5.2.1.3, the removed fittings preserved as specified in 5.2.2, the accessories (except for the required lengths of wire-reinforced hose assemblies), preserved as specified in 5.2.3, and the technical publications, when required, preserved as specified in 5.2.4, shall be consolidated together, as practicable, in one or more snug-fitting boxes conforming to PPP-B-601, domestic type, style A with unnailed top. Cushioning shall be positioned within the box to prevent free movement and damage to the contents. Box closure shall be as specified in PPP-B-601 for the unnailed closure feature. The box shall be legibly marked as follows:

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EMERGENCY REPAIR ITEM
QUICK-DISCONNECT FITTINGS
ACCESSORIES (when provided)
FOR
COLLAPSIBLE, PETROLEUM TANKS

5.2.6 Tank. All openings into the tank shall be sealed with the respective plugs or caps provided for that purpose. Each tank shall be dusted with a talc or other anti-sticking compound to prevent adhesion to itself. Permanently attached fittings shall be protected with cushioning material conforming to and secured in place as specified in 5.2.2. The tank shall be laid flat, completely collapsed, and then folded or rolled to form a neat compact bundle. The bundle shall be secured with web ties. The means of securing the bundle shall not subject the tank to the hazards of any sharp metal objects utilized either as a component of a securing device or as a tool for releasing the means.

5.2.7 Ground cloth. Each ground cloth shall be dusted, folded, or rolled and secured as specified in 5.2.6 for the tank.

5.2.8 Urethane-coated fabrics. Tanks, and as applicable ground cloths, which are made of urethane-coated fabrics, shall be preserved as specified in 5.2.6 and 5.2.7, respectively. Each rolled-up tank shall then be sealed in a water-vaporproof bag in accordance with MIL-P-116, submethod IIc. Each rolled-up ground cloth shall be sealed in a water-vaporproof bag in the same manner as the tank.

5.3 Packing. Each complete tank, and ground cloth when required, preserved as specified in 5.2, shall be packed in accordance with the requirements for level A, B, or C as specified (see 6.2).

5.3.1 Level A. When level A packing is specified (see 6.2), the following special consideration shall be afforded this product because of an anticipated extended period of storage under favorite environmental conditions.

5.3.1.1 Wood preservation (see 6.9). All lumber and plywood used in the construction of the exterior crates/boxes shall be treated with wood preservative in conformance with composition C or D of TT-W-572. Alternatively, the lumber/plywood of the exterior crates/boxes may be treated with the water-borne preservative Chromated Copper Arsenate (CCA) in accordance with processes described in AWPA (American Wood Preservers' Association) Standards C1, C2, and C9. When CCA is used, the following additional conditions must be satisfied.

a. Such treatment shall result in a minimum retention of 0.40 pounds of preservative per cubic foot of lumber/plywood as defined in the applicable standard(s) referenced in C1, C2, and C9.
b. After treatment, the material shall be dried to a moisture content of 19 percent or less.
c. All fasteners used in the fabrication of any component of a wooden container, all fasteners used in the blocking, bracing and anchoring of the contents to any component of a wooden container, and all fasteners used to assemble a wooden container together, shall be hot-dip galvanized fasteners.

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d. All metal surfaces of the product that can come in direct contact with the treated lumber/plywood shall be protected with noncorrosive barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall completely cover the contact surfaces and shall be secured in place with tape conforming to PPP-T-60, type IV, or MIL-T-22085, type II.

5.3.1.2 Exterior container/shipping containers. Exterior container/shipping containers shall be snug-fitting boxes in accordance with the following requirements. The exterior containers/shipping containers shall accommodate a field repack of the tank, ground cloth, and all other accessories and components originally packed with the tank (see 6.11). Carriage bolts, lag screws/bolts, nuts, and washers shall conform to the following:

CARRIAGE BOLTS	= FF-B-584, type I, class 1, style A.
LAG SCREWS/BOLTS	= FF-B-561, type I, grade B
NUTS	= FF-N-836, type I or II, style 1 or 4
WASHERS	= FF-W-92, type A, grade I, class A.

5.3.1.2.1 Plywood. All plywood shall be in accordance with NN-P-530, group B. The plywood shall be interior type, bonded with exterior (waterproof) glue, grade CD.

5.3.1.2.2 Cleats. All lumber for the cleats shall comply with the quality requirements of MIL-STD-731 for class 2 - Structural (members with moderate stress), except that wane is not acceptable. The cleat sizes shall be nominal 1 inch x 4 inches and nominal 2 inches x 4 inches with minimum dimensions for each as shown therein.

5.3.1.2.3 Skids and, as applicable, rubbing strips. All lumber for the skids and, as applicable the rubbing strips, shall comply with the quality requirements of MIL-STD-731 for classes 2 - Structural (members with moderate stress), except the wane is not acceptable. Skids shall be nominal 4 inches x 4 inches; the rubbing strips, when required, shall be nominal 4 inches x 4 inches or nominal 4 inches x 5 inches with minimum dimensions for each as shown therein. Group I wood species shall not be used for the skids.

5.3.1.2.4 Fabrication. Each panel of the shipping container shall be double sheathed with plywood. The inside plywood shall, as specified, be either 0.25 inches thick or 0.375 inches thick and the outside sheathing shall be 0.50 inches thick. The outside sheathing shall be attached to the applicable cleats with the fasteners driven through the C side of the plywood and clinched on the exposed side of the cleat. The inside sheathing shall then be attached to the panel with the fasteners driven through the wood parts in the sequence of the C side of the 0.25/0.375-inch plywood, to cleat, to 0.50-inch plywood. The sizes of the fasteners, the positioning of the fasteners, and the clinching of the fasteners shall be as specified in PPP-B-601; however, care shall be exercised so that the positioning of such fasteners does not interfere with the lag screws/bolts and carriage bolts required to assemble the top, sides, ends and bottom panels together. Clinching of the fasteners shall not be permitted on any inside surface of the container panels.

5.3.1.2.4.1 Side panels. The side panels shall utilize nominal 1-inch x 4-inch lumber for the cleats (see 5.3.1.2.2).

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5.3.1.2.4.2 Top, end, and bottom panels. The top, end, and bottom panels shall utilize nominal 2-inch x 4-inch lumber for the cleats (see 5.3.1.2.2).

5.3.1.2.5 Anti-abrasion provisions. To prevent abrasion, all interior surfaces of the shipping container that will contact the tank, and as applicable the ground cloth, including any applicable outer surface of the consolidation containers(s), shall be lined with a layer of scrap tank fabric, cushioning material conforming to PPP-C-1797, or polyethylene film conforming to L-P-378, type I, class 1, with a minimum thickness of 6 mils. The material shall be applied in such a manner that it does not interfere with the removal of the container top, sides, and ends as a joined one-piece assembly or as individual panels. The means of attaching the material shall not create a hazard within the container that could possibly damage the contents. As an alternative, each tank, and ground cloth, when required, preserved as required in 5.2.6 through 5.2.8, shall be individually loosely wrapped with the anti-abrasion material. The manner of wrapping shall provide a minimum of one layer of material around the tank, and ground cloth when required, to prevent direct contact with other components and the inside surfaces of the shipping container.

5.3.1.2.6 Filled consolidation container(s). The filled consolidation containers(s) shall be positioned within the shipping container in a manner to prevent movement. The positioning method shall utilize the shipping container bottom only; it shall not be dependent upon the application of devices of the sides, top, or ends of the container.

5.3.1.2.7 Wire-reinforced hose assemblies. The wire-reinforced hose assemblies, preserved as specified in 5.2.3.1, shall be positioned and secured within the shipping container in such a manner to ensure that the top, sides, and ends of a shipping container may be removed from the container bottom while the hoses remain secured in place without damage to the tank, and as applicable, the ground cloth and water-vaporproof bag specified in 5.2.8.

5.3.1.2.8 Containers with an outside length dimension of 96 inches or less. Containers with an outside length dimension of 96 inches or less shall be a snug-fitting box conforming to PPP-B-601, overseas type, style A, type 3 load, except that each panel shall be double sheathed with the cleats sandwiched between two sheets of plywood (see 5.3.1.2.4). The inside plywood shall be 0.25 inches thick and the outside plywood shall be 0.50 inches thick. Other exceptions are as follows:

5.3.1.2.8.1 Side panels. The positioning of the edge cleats, and intermediate cleats when required, shall be as specified for the sides in PPP-B-601. The size of the side panel shall be such that its length (the longest dimension) is the same as the outside length of the box. The width of the side panel shall be such that it is the same as the inside depth of the box plus the thickness of the double sheathed bottom panel (see figure 18).

5.3.1.2.8.2 Top panel. The position of the edge cleats, and intermediate cleats when required, shall be as specified for the top in PPP-B-601. The size of the top panel shall be the same as specified therein for style A; that is, the length and width of the panel shall be the same as the outside length and outside width of the box (see figures 18 and 19).

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5.3.1.2.8.3 End panels. The positioning of the edge cleats, and intermediate cleats when required, shall be as specified for the ends in PPP-B-601. The size of the end panel shall be the same as specified therein for style A; that is, the length and width shall be the same as the inside width and inside depth of the box (see figure 18).

5.3.1.2.8.4 Bottom panel. The positioning of the edge cleats, and intermediate cleats when required, shall be as specified for the bottom in PPP-B-601 except that the thru-edge cleats shall run parallel with the box length and such cleats shall be the same length as the outside length of the box. The size of the bottom panel shall be as specified therein for style A except that the width of the bottom panel shall be the same as the inside width of the box (see figure 18).

5.3.1.2.8.5 Skids. Skids (see 5.3.1.2.3) shall be applied to the bottom panel of each box, running parallel with and extending the full outside width of the box. A bevel 45 degrees (±5 degrees) shall be applied to the bottom third portion of the skid ends. The skids shall be set back from the end of the box a distance of not less than 3 inches or more than 1/6 of the box length. Each box shall be provided with a minimum of two skids with additional skids being provided when the distance between the inside edges of the skids exceeds 48 inches. The positioning of the additional skids shall divide the area between the end skids into units of equal space. Additional intermediate cleats shall be incorporated into the bottom panel at time of fabrication to assist in the attachment of the skids and to offer full support to the skids. Each skid shall be fastened to the bottom panel with 0.375-inch diameter carriage bolts that are 5.25 inches long (+0.00 inches, -0.25 inches). The bottom panel shall be positioned so that it is set back from the ends of the skids a distance that is equal to the thickness of the box side panels. The holes for the carriage bolts shall be the exact diameter of the bolt. Each hole shall be counterbored on the underside of the skid so that 3 full threads (+3, -0) extend beyond the washer and nut when the nut has been properly tightened. The nut shall be tightened so that the bolt head is drawn into the plywood and cleat to prevent the bolt head from becoming a snagging hazard on the inside of the box; however, no portion shall extend beyond the bottom of the skid when tightened. After tightening the nuts, the threads of the bolt extending beyond the nut shall be painted with a suitable metal primer or similar material. The bolts shall be centered at 6.50 inches (±0.25 inches) from each end of the skid with additional bolts positioned in a staggered pattern, as practicable, so that the distance between their centerlines shall be no more than 16 inches (±0.25 inches) and no less than 8 inches (±0.25 inches) (see figure 21). Each skid shall be slotted/notched sufficiently to provide clearances for strapping. Figures 18, 19, and 21 give details of skid application.

5.3.1.2.8.6 Assembly. The box panels shall be joined together with lag screws/bolts and flat washers. Entry holes and lead holes shall be provided in each box panel for proper assembly as follows:

5.3.1.2.8.6.1 Entry holes. Entry holes, the same size as the lag screws, shall be provided through each applicable flat panel surface. The holes shall be set back from the edges of that flat surface to provide the edge distance necessary for the leg screws to be centered on the nominal 1-inch, or as applicable the nominal 2-inch, cleat thickness of the joining panel. The entry holes for the 0.25-inch diameter lag screw for joining a panel to a nominal

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1-inch thick cleat shall be centered at 0.875 inches from the edge of the flat panel. The entry holes for the 0.375-inch diameter lag screw for joining a panel to a nominal 2-inch thick cleat shall be centered at 1.25 inches from the edge of the flat surface (see figures 18, 19, 20, and 21).

5.3.1.2.8.6.2 Lead holes. Lead holes for the threaded portion of the lag screw shall be provided in each applicable panel edge. The holes shall be drilled only to the depth necessary to accommodate the penetration of the threaded portion of the lag screw. The lead holes for the 0.25-inch diameter lag screw shall be 0.1875 inches in diameter. The lead holes for the 0.375-inch diameter lag screws shall be 0.25 inches in diameter. The lead holes shall provide the same edge distance as specified in 5.3.1.2.8.6.1 for the entry holes.

5.3.1.2.8.6.3 Lag screw application. The lag screws shall be turned in their holes the full length of the screw. They shall not be driven in by hammer or by any other similar means. If for any reason the thread in the wood is stripped when the lag screws are placed, the lag screw shall be removed and placed in a new hole near the old position. A single flat washer shall be used under the head of each screw.

5.3.1.2.8.6.4 Fastening bottom assembly to end panels. The bottom assembly shall be fastened to each end panel with a minimum of four lag screws 0.375 inches in diameter by 5 inches in length. The positioning of each screw shall be as shown in figure 18.

5.3.1.2.8.6.5 Fastening side panels to end panels. The side panels shall be fastened to the end panels with 0.375-inch diameter lag screws that are 4 inches long. The screws shall be positioned as shown in figure 19.

5.3.1.2.8.6.6 Fastening of side panels to bottom assembly. The side panels shall be fastened to the bottom assembly with 0.375-inch diameter lag screws that are 4 inches long. The screws shall be positioned as shown in figure 19.

5.3.1.2.8.6.7 Fastening top panel to side panels. The top panel shall be fastened to the side panels with 0.25-inch diameter lag screws that are 5 inches long. The screws shall be positioned as shown in figure 20.

5.3.1.2.8.6.8 Fastening top panel to end panels. The top panel shall be fastened to the end panels with 0.375-inch diameter lag screws that are 5 inches long. The screws shall be positioned as shown in figure 20.

5.3.1.2.8.6.9 Strapping. Strapping of the assembled box shall be as specified in the appendix to PPP-B-601 except that the strapping shall be flat and the finish shall be "B".

5.3.1.2.9 Container with an outside dimension greater than 96 inches. Containers with an outside length dimension greater than 96 inches shall be a snug-fitting box conforming to MIL-B-26195, type II, style A, class 1, with full plywood panels for the base, except that each panel of the box shall be double sheathed with the cleats sandwiched between two sheets of plywood (see 5.3.1.2.4). The outside plywood shall be 0.50 inches in thickness. The inside plywood shall be 0.25 inches thick for the sides and ends, and shall be 0.375 inches thick for the top and bottom. Other exceptions are as follows:

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5.3.1.2.9.1 Side panels. The positioning of the edge cleats and intermediate cleats shall be as specified for the sides in MIL-B-26195, that is the length of the thru-edge cleats shall be the same as the outside length of the box. The outside plywood sheathing shall extend above the upper thru-edge cleat a distance equal to the thickness of the cleats and inside plywood sheathing of the top panel (see figure 22 for the end view of box and figure 26 for assembly method).

5.3.1.2.9.2 Top panel. Thru-edge cleats and thru-intermediate cleats shall run parallel with the width of the box. The size of the outside plywood shall be the same as the outside length dimension and the outside width dimension of the box. The inside plywood shall have the same length as the outside length of the box but shall have a width that is 1 inch less than the outside box width. The length of the thru-edge cleats and thru-intermediate cleats shall be the same as the width of the inside plywood and shall be so positioned to be flush with the edges of the inside plywood. Filler-edge cleats shall run parallel with the box length and shall be positioned flush with the side edges of the inside plywood (see figures 22, 23, 26, and 27).

5.3.1.2.9.3 End panels. The positioning of the cleats shall be as specified in MIL-B-26195.

5.3.1.2.9.4 Base. The base shall consist of a deck assembly to which the skids and rubbing strips shall be attached. The deck shall be double sheathed with 0.375-inch thick plywood as the upper (inner) side and 0.50-inch thick plywood as the outside. The headers and load-bearing numbers shall be nominal 2-inch x 4-inch lumber and shall be utilized as thru-edge cleats and thru-intermediate cleats positioned to run parallel with the box width dimension. The filler-edge cleats shall run parallel with the box length, shall be positioned between the thru cleats flush with the edge of the plywood and shall be nominal 2-inch x 4-inch lumber. However, filler cleats shall not be required in those spaces where the distance between thru cleats is 4 inches or less. The deck assembly shall be attached to the skids with either of two lengths of carriage bolts. The shorter length shall attach the deck assembly to the skid only. The longer length shall attach the deck assembly to the skids and rubbing strips with a minimum of two fasteners extending through the two longer rubbing strips. The short, 16-inch long, center rubbing strip shall be attached to the deck assembly with a total of two of the longer carriage bolts. When an intermediate skid is required, additional filler cleats shall be positioned above that skid between the thru-edge cleats and the thru-intermediate cleats. Carriage bolts shall be utilized to attach the intermediate skid and rubbing strips to the deck assembly in the manner specified for the outer skids. The length of the deck assembly shall be the same as the inside width of the box. Figures 22, 25, 27, and 28 give details of the base.

5.3.1.2.9.4.1 Skids. A minimum of two skids of nominal 4-inch x 4-inch lumber shall be attached to the underside of the deck assembly in direct contact with the 0.50-inch thick plywood sheathing. An intermediate skid of the same size and species of lumber shall be required on all boxes with an outside width in excess of 36 inches. The skids shall be positioned as specified in MIL-B-26195. The 45-degree bevel shall not be cut into the ends of the skid. The bottom, inside corner of each outside skid shall be chamfered full length at a 60-degree angle (±5 degrees) to eliminate a potential snag when attempting to lift from the end with a forklift truck (see figures 22 and 26). Intermediate skids shall not be chamfered.

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5.3.1.2.9.4.2 Rubbing strips. The rubbing strips shall be nominal 4 inches x 5 inches for the outside skids and, when required, nominal 4 inches x 4 inches for the intermediate skid. The rubbing strips shall be beveled full depth at an angle of 45 degrees (±5 degrees) at sling and forklift truck openings. The rubbing strips shall be positioned on each skid to provide two 12-inch wide openings for forklift truck access. Such openings shall be spaced 28 inches apart, center to center, and shall be positioned to straddle the center of balance of the loaded box. The sling openings at the ends of each skid shall be 8 inches in length. Each nominal 4-inch x 5-inch rubbing strip shall be attached to the base in a manner to provide a ledge for supporting the full thickness of the box side panel (see figures 22 and 26).

5.3.1.2.9.4.3 Base fabrication. The skids and rubbing strips shall be attached to the deck assembly as specified in 5.3.1.2.9.4.1 and 5.3.1.2.9.4.2 and as shown in figures 25, 26, 27 and 28. The thru-edge cleats and thru-intermediate cleats of the deck assembly shall be positioned as shown in figure 28. Such cleats shall be utilized as headers and load-bearing-floor boards, shall serve as the means for locating the carriage bolts, and shall be positioned as shown in figure 28. The carriage bolts shall be 0.375 inches in diameter and the exposed threads after tightening shall be painted as specified in 5.3.1.2.8.5. A single flat washer shall be required under the nut for each carriage bolt. The holes for the carriage bolts shall be the exact diameter of the bolt. Each hole shall be counterbored on the underside of the rubbing strips and, as applicable, the skids so that 3 full threads (+3, -0) extend beyond the washer and nut when properly tightened. The nut shall be tightened to satisfy the requirements of 5.3.1.2.8.5.

5.3.1.2.9.5 Assembly. The box panels and base shall be joined together with lag screws. A flat washer shall be required under the head of each screw.

5.3.1.2.9.5.1 Entry holes. Entry holes, the same size as the lag screw, shall be provided through each applicable flat panel surface. The holes shall be set back from the edges of that flat surface to provide the edge distance necessary for the lag screws to be centered on the nominal cleat thickness of the joining panel (see figures 22, 23 and 24).

5.3.1.2.9.5.2 Lead holes. Lead holes, 0.25 inches in diameter, shall be provided for the threaded portion of each lag screw. The holes shall be drilled only to the depth necessary to accommodate the penetration of the threaded portion of the lag screw. The holes shall be positioned as shown in figures 22, 23, and 24.

5.3.1.2.9.5.3 Lag screw application. Lag screw application shall be as specified in 5.3.1.2.8.6.3.

5.3.1.2.9.5.4 Fastening end panels to base. The end panels shall be fastened to the base with 0.375-inch diameter lag screws that are 5 inches long. The positioning and quality of the screws shall be as shown in figure 22.

5.3.1.2.9.5.5 Fastening of side panels to base and end panels. The side panels shall be fastened to the base and end panels with 0.375-inch diameter lag screws that are 4 inches long. The positioning and quality of the screws shall be as shown in figure 23.

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5.3.1.2.9.5.6 Fastening of side panels to top panel. The side panels shall be fastened to the top panel with 0.375-inch diameter lag screws that are 3 inches in length. The positioning and quantity of the screws shall be as shown in figure 23.

5.3.1.2.9.5.7 Fastening of top panel to end panels. The top panel shall be fastened to the end panels with 0.375-inch diameter lag screws that are 5 inches long. The positioning and quantity of the screws shall be as shown in figure 24.

5.3.1.2.9.5.8 Strapping. Corner straps shall be applied to the assembled box in accordance with MIL-B-26195 except that the finish of the strapping shall be finish B.

5.3.2 Level B. Each complete tank, and ground cloth when required, preserved as specified in 5.2, shall be packed for level B in the same manner as specified for level A in 5.3.1 except that the lumber and plywood preservation requirements of 5.3.1.1 shall be omitted.

5.3.3 Level C. Each complete tank, and ground cloth when required, preserved as specified in 5.2, shall be packed in a close-fitting box conforming to, as applicable to the weight of the contents, PPP-B-601, domestic type for type 3 load, or MIL-B-26195, type I, style C, with plywood panels, class 1, with full panel deck and rubbing strips. The box shall accommodate a field repack of the tank, ground cloth, and all other accessories and components originally packed with the tank (see 6.11). The contents shall be protected against abrasion as specified in 5.3.1.2.5, the filled consolidation container(s) shall be positioned as specified in 5.3.1.2.6, and the wire-reinforced hose assemblies shall be positioned and secured as specified in 5.3.1.2.7. Box closure and strapping shall be as specified in the applicable box specification or the appendix thereto except that strapping shall be flat and the finish may be A.

5.4 Anchoring of tank. The anchoring of the rolled-up tank to the bottom/base of the box shall not be required; however, all contents of the box shall be blocked and braced within the box in a manner to prevent longitudinal movement when testing in accordance with the rail impact of MIL-STD-1186 is required (see 4.7.1.2.1).

5.5 Marking. In addition to the special markings specified in 5.5.1 and any special or identification markings required by the contractor or purchase/delivery order, each container shall be marked in accordance with MIL-STD-129, including all appendices, and as applicable to overseas shipment. Special care must be exercised to ensure the following:

a. That interior and exterior packing lists are applied as specified.
b. That marking surfaces are prepared as specified.
c. That all markings are overcoated with waterproof coating as specified.
d. That when the tanks are to be included as part of a set, assembly, or module, each container is marked as specified.

5.5.1 Additional special markings. Additional special markings shall be applied to the outside of the box as follows:

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5.5.1.1 Marking for unpacking. Additional special markings shall be applied to the outside of the box to show the "FLOW DIRECTION" of the deployed tank relative to the folded and rolled-up tank as it is positioned within the box. Additionally, markings shall be applied to the outside of the box listing in sequence the operations required to properly remove the top, sides, and ends from the base as individual components, or as a one-piece assembly. Alternatively, such unpacking instruction may be as preprinted matter to be included with the packing list in the exterior packing list envelope. When the unpacking instructions are placed in the envelope, a suitable area immediately adjacent to the envelope shall be marked with the words "UNPACKING INSTRUCTIONS INSIDE PACKING LIST ENVELOPE," with an arrow pointing to that envelope. Any code utilized in the instructions to identify critical locations on the box shall be marked on the outside of the box at that location with the corresponding code.

Special Note: The design of the box with an outside length of 96 inches or less for level A is such that the proper manner for removing the end panels from the base or bottom is by removing the lag screw from the underside of the box (see 1 on figure 18).

5.5.1.2 Marking for handling of treated wood. Each box constructed of treated lumber and plywood, shall, as applicable, be marked on an exterior surface in a prominent position with instructions advising involved personnel of any precautions to be exercised in handling of the treated lumber. Such instructions shall include recommended limitations in the use and disposal of the treated lumber and shall be in the format developed by the treated wood industry's Consumer Awareness Program (CAP) as approved by the Environmental Protection Agency (EPA).

5.5.1.2.1 Methods of marking. The instructions required in 5.5.1.2 shall be applied to the shipping container in the manner specified in MIL-STD-129 for stenciling or labeling.

5.5.1.2.1.1 Alternate method. As an alternate, such instructions may be as preprinted matter to be included with the packing list in the exterior packing list envelope. When the instructions are placed in the envelope, a suitable area immediately adjacent to the envelope shall be marked with the words "INSTRUCTIONS FOR HANDLING TREATED WOOD INSIDE PACKING LIST ENVELOPE", with an arrow pointing to that envelope.

5.5.1.2.2 Sample instructions. An example of an EPA approved set of instructions for the proper handling of CCA (see 5.3.1.1) is shown in figure 29. Other formats may be utilized provided the instructions included in the format are EPA approved.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The collapsible tank is intended for use as a fuel storage container when quick storage facilities are needed, and where permanent fuel storage facilitates are not available, or when the storage of fuel is needed only on a temporary basis. It is intended that the collapsible tank provide a 10-year shelf life and 5-year use life.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Capacity of tank required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is not required for inspection and approval (see 3.2).
e. Time frame for submission of first article pack (see 5.1).
f. When ground cloth is not required (see 3.7).
g. When accessories are not required (see 3.11).
h. Type of fuel to be utilized for fuel storage and overload tests when contractor's first article is specified (see 4.5.2.6 and 4.5.2.7).
i. Level of packing required (see 5.3).
j. Any special marking (see 5.5).
k. Part or identifying number required (see 3.12 and 6.13).
l. When additional first article samples are not required (see 3.2.3).

6.3 First article. When a first article inspection is required, the item should be a first production item. The first article should consist of two or more units. In addition, the first article must include at least one type repair for validation of each repair procedure. The repairs may be applied to the first tanks or to separate samples. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.3.1 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to preserve, pack, and mark the tanks in accordance with this specification.

6.4 Sampling selection plan. The contractor should note that the sample selection plan (see 3.2.1) and process data (see 3.3) require Government approval prior to construction of the first article tank.

6.5 Nylon solution. The nylon material elvamide 8061, manufacturer's code 18873, was found to meet the performance requirements and is suggested for use as a guide only. The Government assumes no responsibility for material or solution thereof.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of tanks.

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6.7 Subject term (key word) listing.

Coated fabric
Collapsible fabric tanks
Elastomeric tanks
Fittings
Petroleum
Rubberized tanks

6.8 Definitions.

6.8.1 Rework. Rework shall be defined as an operation performed during a production operation due to an inadequacy or error during that operation.

6.8.2 Repair. Repair shall be defined as a corrective operation which is required because of an omission of a step or incorrect performance of a step or process during a previously completed manufacturing operation.

6.9 Wood preservative (see 5.3.1.1). The intent of the wood preservative applications specified in 5.3.1.1 is to prolong the serviceability of the wooden shipping containers. Compositions C and D of TT-W-572 can offer reliable protection for periods up to 30 years against decay and wood-destroying organisms provided such treated containers are not stored in direct contact with the ground. The preservative CCA provides reliable protection against the same undesirable elements for periods as long as 40 years. However, the properties of CCA and the prescribed methods of application are such that the treated container may be stored in direct contact with the ground, even buried underground, in the 40-year period and continue to provide the desired degree of reliable protection. When level A packing is a requirement of the contract and it is known that the containers are to be exposed to unfavorable conditions for an undetermined period of time, the contracting officer should request the use of the wood preservative CCA.

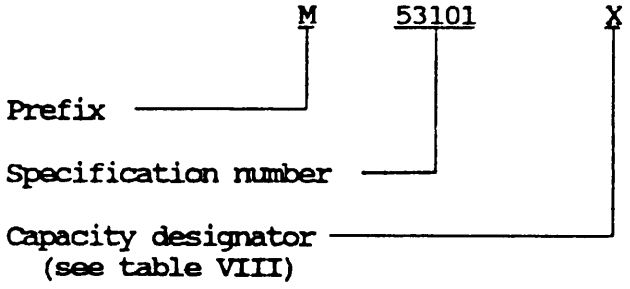
6.10 Fabric for handles and chafing patches. Coated fabric which meets the requirements of table II and the color requirements of this specification may be used for handles and chafing patches. Different color patches may be used for internal chafing patches only.

6.11 Field repack. The tank container is intended to accommodate a field repack. As a guide only, it is suggested that the tank and accessories would require an increase of five percent length and width and ten percent in depth for repack after use in the field. Manufacturers should consider that the tank will not be folded as tightly after field use as when folded for packaging at the manufacturing facility.

6.12 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

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6.13 Part or identifying number (PIN). The PIN to be used for tanks acquired to this specification are created as follows (see 3.12 and 6.2):



6.13.1 Capacity designator. The capacity designator is a two-position field used to designate the required tank capacity (see table VIII)

TABLE VIII. Capacity designator.

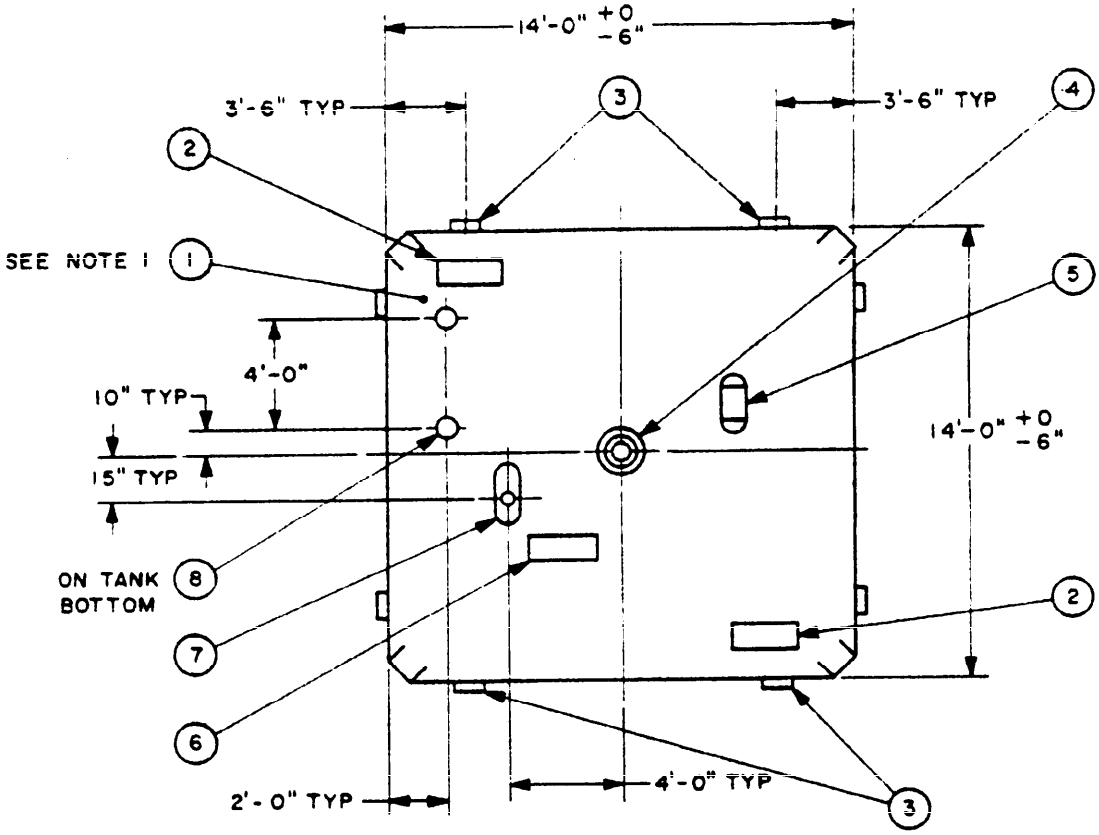
Capacity designator	Remarks
03	3,000 gallon
10	10,000 gallon
20	20,000 gallon

Custodians:
Army - ME

Preparing activity:
Army - ME

Project 5430-0186

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TOP VIEW

NOTES:

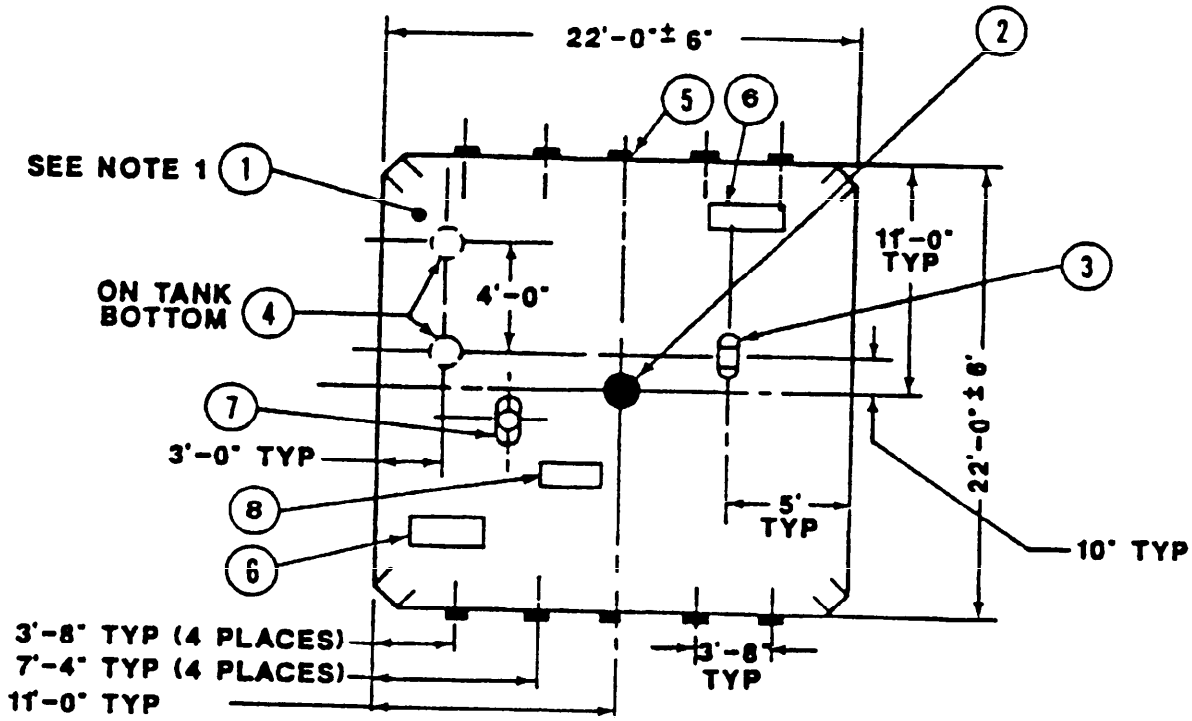
1. TANK IN EMPTY CONFIGURATION
2. APPROXIMATE DIMENSIONS OF FILLED TANK: 12'-6" X 12'-6" X 4'-0" HIGH.
3. SHAPE OF TANK CORNERS OPTIONAL.
4. FOLLOWING SIZE CHAFING PATCHES SHALL BE BONDED INSIDE AND OUTSIDE THE TANK:
   OPPOSITE ITEM 2 -18±4 IN. DIA OR SQ.
   OPPOSITE ITEM 3 AND 8 -36±4 IN. DIA OR SQ.
   OPPOSITE ITEM 5 -18±4 IN. DIA OR SQ.
5. ALL TOLERANCES ±1 FOOT UNLESS OTHERWISE NOTED.
6. GENERAL LOCATION OF LABELS IS SHOWN. EXACT ORIENTATION IS AT MANUFACTURER'S DISCRETION.

ITEM	QTY	DESCRIPTION
1	1	TANK, FABRIC, COLLAPSIBLE, 3,000 GAL
2	2	ID LABELS
3	8	TANK HANDLES
4	1	VENT PIPE ASSEMBLY W/PRESSURE RELIEF CAP
5	1	DISCHARGE ASSEMBLY
6	1	CAUTION LABEL
7	1	FILLER ASSEMBLY
8	2	DRAIN FITTING

FIGURE 1. Tank fabric, collapsible, 3,000 gal.

X-4851

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TOP VIEW

NOTES:

1. TANK IN EMPTY CONFIGURATION.
2. APPROXIMATE DIMENSIONS OF FILLED TANK: 20' - 6" x 20' - 6" x 4' - 0" HIGH.
3. SHAPE OF TANK CORNERS OPTIONAL.
4. FOLLOWING SIZE CHAFING PATCHES SHALL BE BONDED INSIDE AND OUTSIDE TANK:

   OPPOSITE ITEM (2) -18 ± 4 IN. DIA OR SQ.

   OPPOSITE ITEM (3)(7) -36 ± 4 IN. DIA OR SQ.

   OPPOSITE ITEM (4) -18 ± 4 IN. DIA OR SQ.

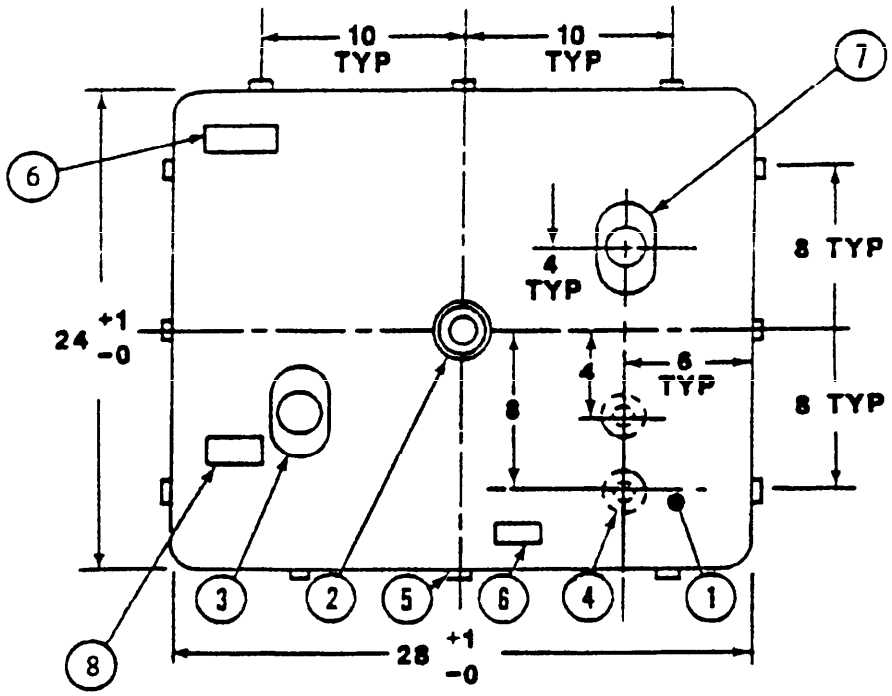
5. ALL TOLERANCES ± 1 FOOT UNLESS OTHERWISE NOTED.

ITEM	QTY	DESCRIPTION
(1)	1	TANK, FABRIC, COLLAPSIBLE, 10,000 GAL.
(2)	1	VENT PIPE ASSEMBLY w/ PRESSURE RELIEF CAP.
(3)	1	DISCHARGE ASSEMBLY
(4)	2	DRAIN FITTING ASSEMBLY.
(5)	10	TANK HANDLE.
(6)	2	I.D. LABELS.
(7)	1	FILLER ASSEMBLY
(8)	1	CAUTION LABEL.

FIGURE 2. Tank, fabric, collapsible, 10,000 gallon petroleum.

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REQUIRED DIMENSIONS OF EMPTY TANK

NOTES

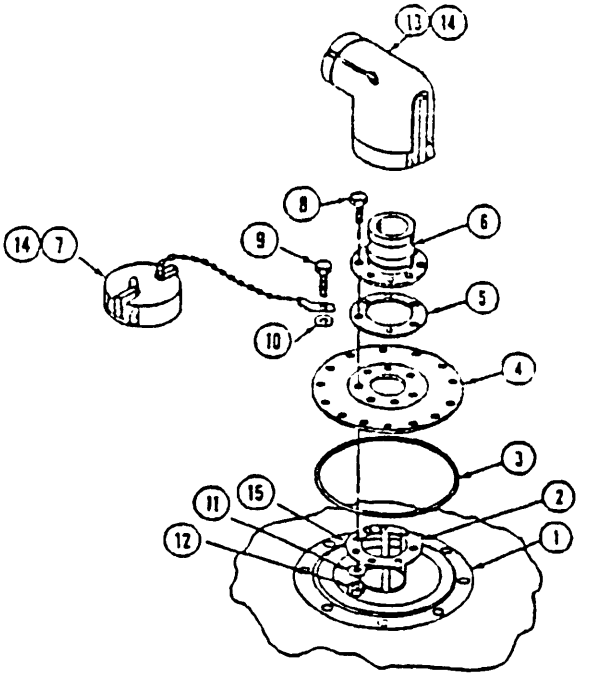
1. ALL DIMENSIONS IN FEET
2. ALL TOLERANCES ± 1 FOOT UNLESS OTHERWISE NOTED
3. SHAPE OF CORNERS OPTIONAL
4. THE FOLLOWING SIZE CHAFING PATCHES SHALL BE BONDED INSIDE AND OUTSIDE THE TANK OPPOSITE EACH FITTING:
   - ITEM (2) -18 ± 4 IN. DIA. OR SQUARE
   - ITEM (3) (7) -36 ± 4 IN. DIA. OR SQUARE
   - ITEM (4) -18 ± 4 IN. DIA. OR SQUARE

ITEM	QTY	DESCRIPTION
(1)	1	TANK 20,000 GAL.
(2)	1	VENT ASSY
(3)	1	FILLER ASSY
(4)	2	DRAIN ASSY
(5)	12	HANDLE
(6)	2	I.D LABELS
(7)	1	DISCHARGE ASSY
(8)	1	CAUTION LABEL

FIGURE 3. Tank, fabric collapsible, 20,000 gal. petroleum.

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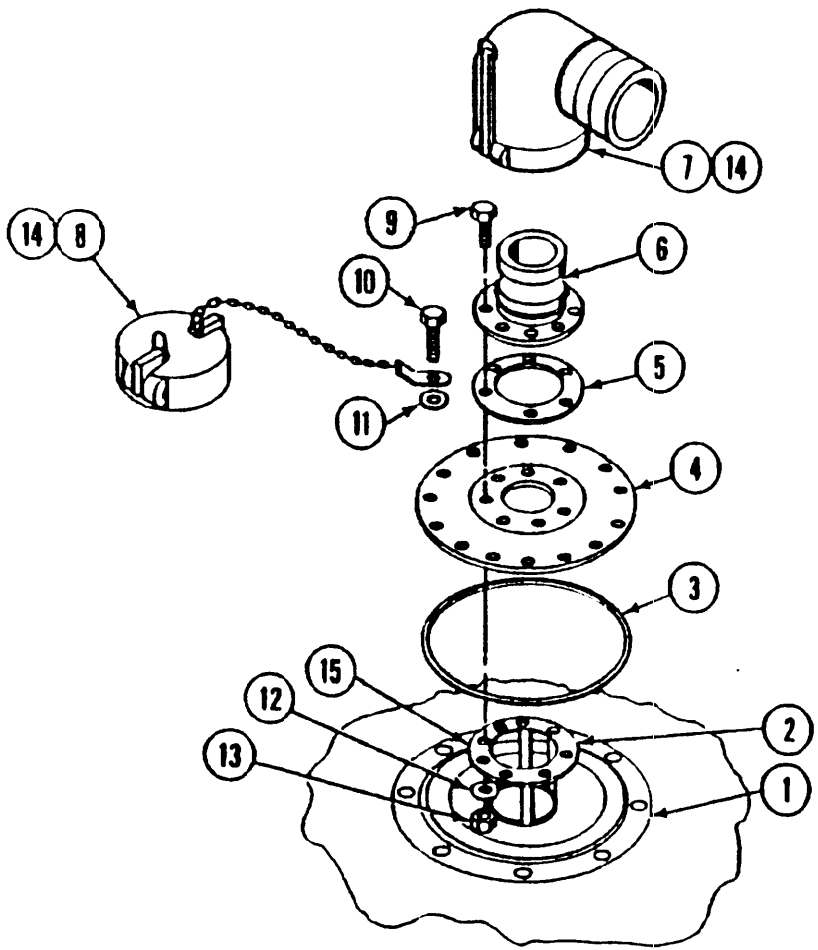
FIND NO.	PART NO. IDENTIFICATION NO.	QTY FILLER FITTING ASSY	DESCRIPTION
1	FIGURE 6	1	ACCESS DOOR, FITTING COMP TYPE
2	FIGURE 9	1	SUCTION STUB, 4 INCH
3	M25988/1-383	1	"O" RING GASKET, MIL-R-25988/1, MIL-R-25988
4	FIGURE 7	1	OVAL CLOSURE PLATE, COMP. FITTING
5	13228E1827-4	1	GASKET
6	MS27023-17	1	COUPLING HALF, FLANGED, 4 IN SIZE
7	MS27028-17	1	DUST CAP, 4 IN, W/GASKET AND SECURITY CHAIN
8	MS90728-64	8	SCREW, CAP, HEX HD., .375-16 UNC X 1.50 L
9	MS90728-8	20	SCREW, CAP, HEX HD., .25-20 UNC X 1.00 L
10	MS27183-10	20	WASHER, PLAIN, 1/4 IN ID
11	MS35338-46	8	WASHER, LOCK, HELICAL, .375 SIZE
12	MS35649-2382	8	NUT, HEX, MACH SCREW, .375-16 UNC
13	81718/633-4	1	ELBOW, FEMALE TO FEMALE, 4 IN X 90° SIZE
14	13228E1768-9	2	GASKET, COUPLING HALF, 4 IN SIZE
15	MAS1523-6R	8	PACKING WITH RETAINER, .375 SIZE

NOTES:

1. LUBRICATE "O" RING WITH SILICONE COMPOUND PER MIL-S-8660 BEFORE ASSEMBLING HARDWARE.

FIGURE 4. Filler assembly, 3,000, 10,000 and 20,000 gal. tanks.

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FIND NO.	PART NO. IDENTIFICATION NO.	QTY FILLER FITTING ASSY	DESCRIPTION
1	FIGURE 6	1	ACCESS DOOR, FITTING COMP TYPE
2	FIGURE 10	1	SUCTION STUB, 6 INCH
3	M25988/1-383	1	"O" RING GASKET, MIL-R-25988/1, MIL-R-25988
4	FIGURE 8	1	OVAL CLOSURE PLATE, COMP FITTING
5	13228E1827-5	1	GASKET
6	MS27023-19	1	COUPLING HALF,FLANGED, 6 IN SIZE
7	81718/633-K-6	1	ELBOW FEMALE TO MALE, 6 IN 90°
8	MS27028-19	1	DUST CAP 6 IN W/GASKET AND SECURITY CHAIN
9	MS90728-64	12	SCREW, CAP, HEX HD. .375-16 UNC X 1.50 L
10	MS90728-8	20	SCREW,CAP,HEX HD,.25-20 UNC X 1.00 L
11	MS27183-10	20	WASHER,PLAIN 1/4 IN ID
12	MS35338-46	12	WASHER,LOCK HELICAL .375 SIZE
13	MS35649-2382	12	NUT,HEX MACH SCREW .375-16 UNC
14	13228E1768-10	2	GASKET-COUPLING HALF, 6 IN SIZE
15	NAS1523-6R	12	PACKING WITH RETAINER .375 SIZE

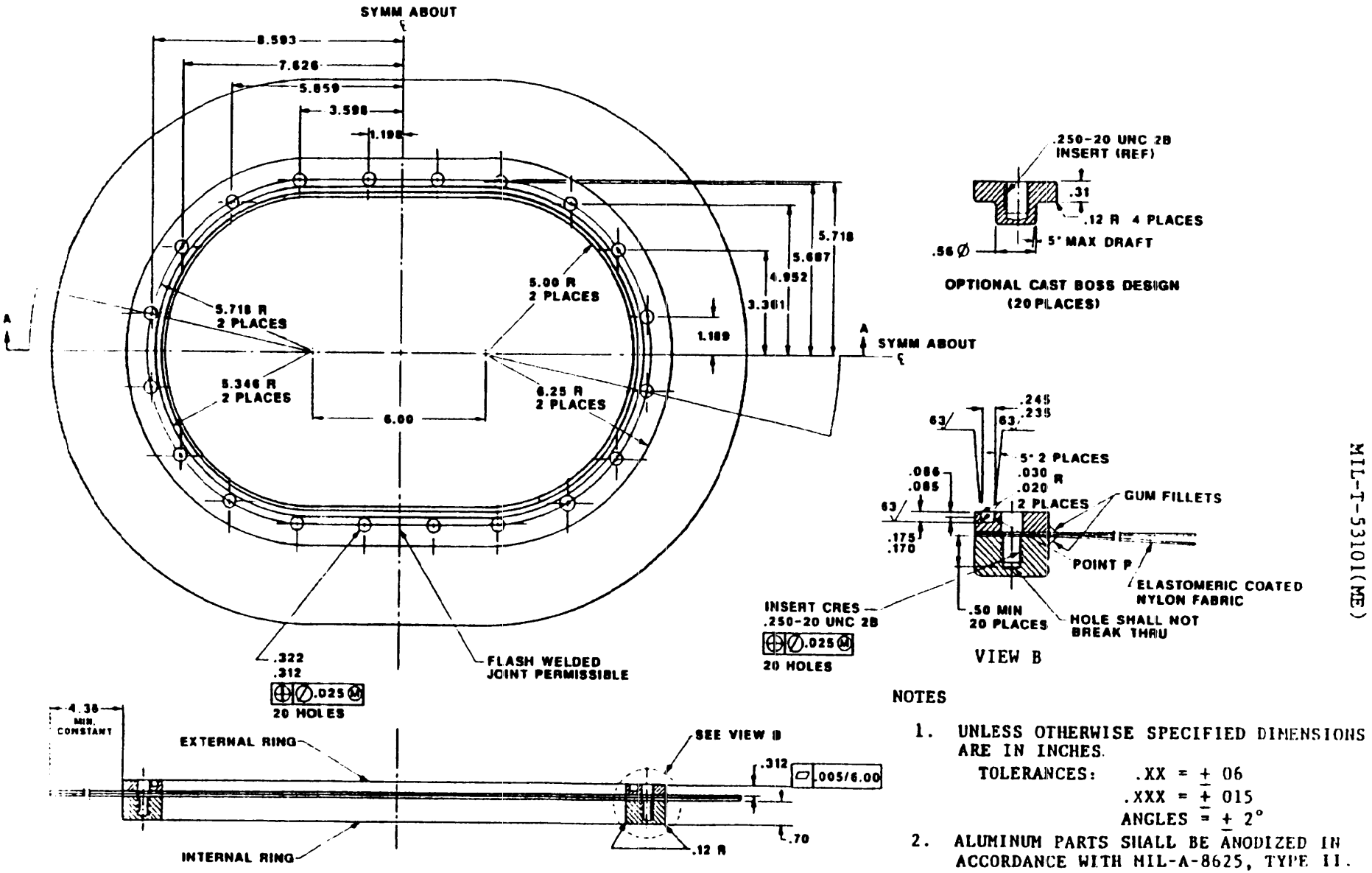
NOTES:

1. LUBRICATE "O" RING WITH SILICONE COMPOUND PER MIL-S-8660 BEFORE ASSEMBLING HARDWARE.

FIGURE 5. Discharge assembly, 3,000, 10,000 and 20,000 gal. tanks.

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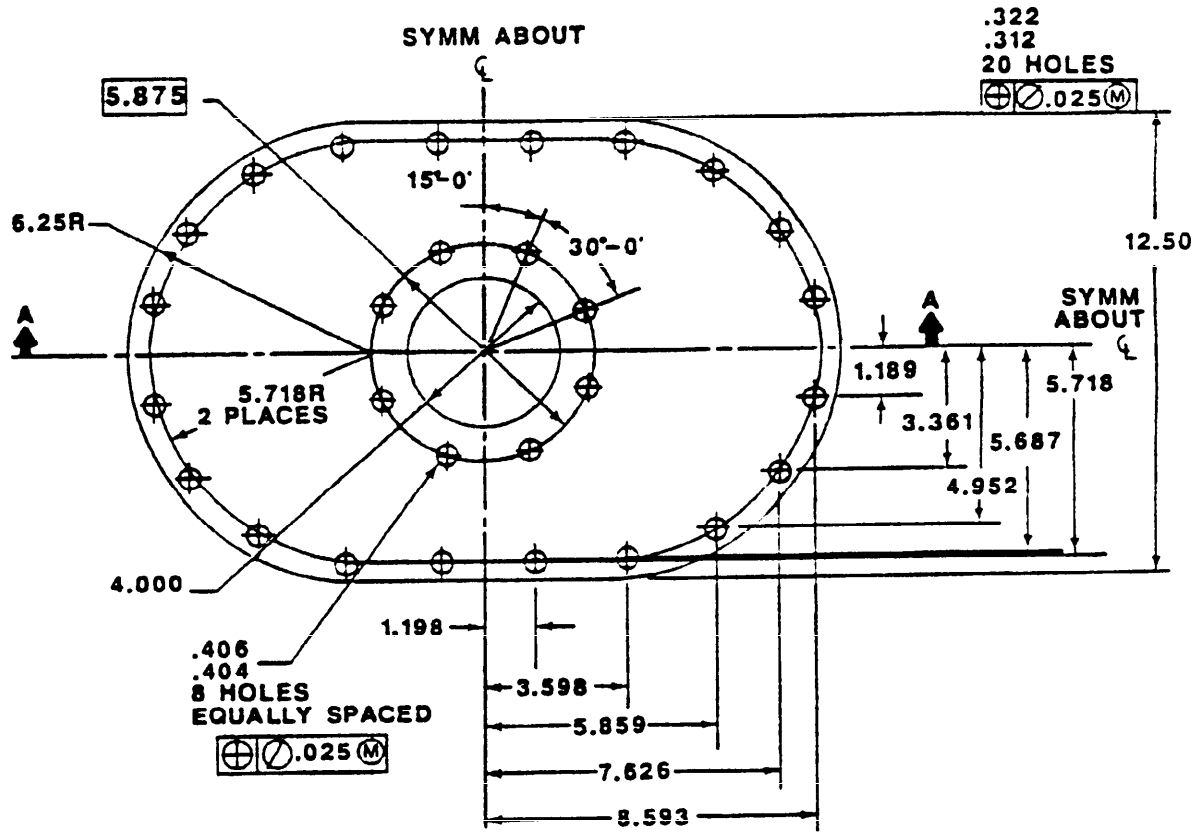
NOTES

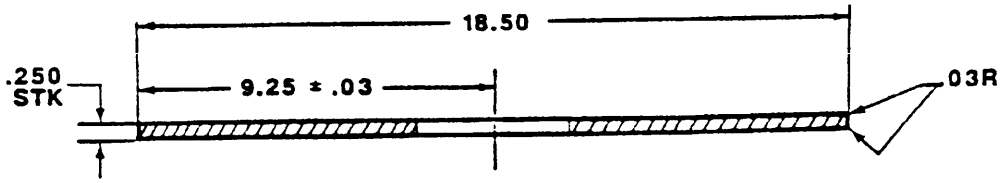
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
   TOLERANCES: .XX = ± 06
   .XXX = ± 015
   ANGLES = ± 2°
2. ALUMINUM PARTS SHALL BE ANODIZED IN ACCORDANCE WITH MIL-A-8625, TYPE II.
3. ADHESIVE BOND FABRIC AND RINGS.

X-4760

FIGURE 6. Access door fi(:g-compression type (

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SECTION A-A

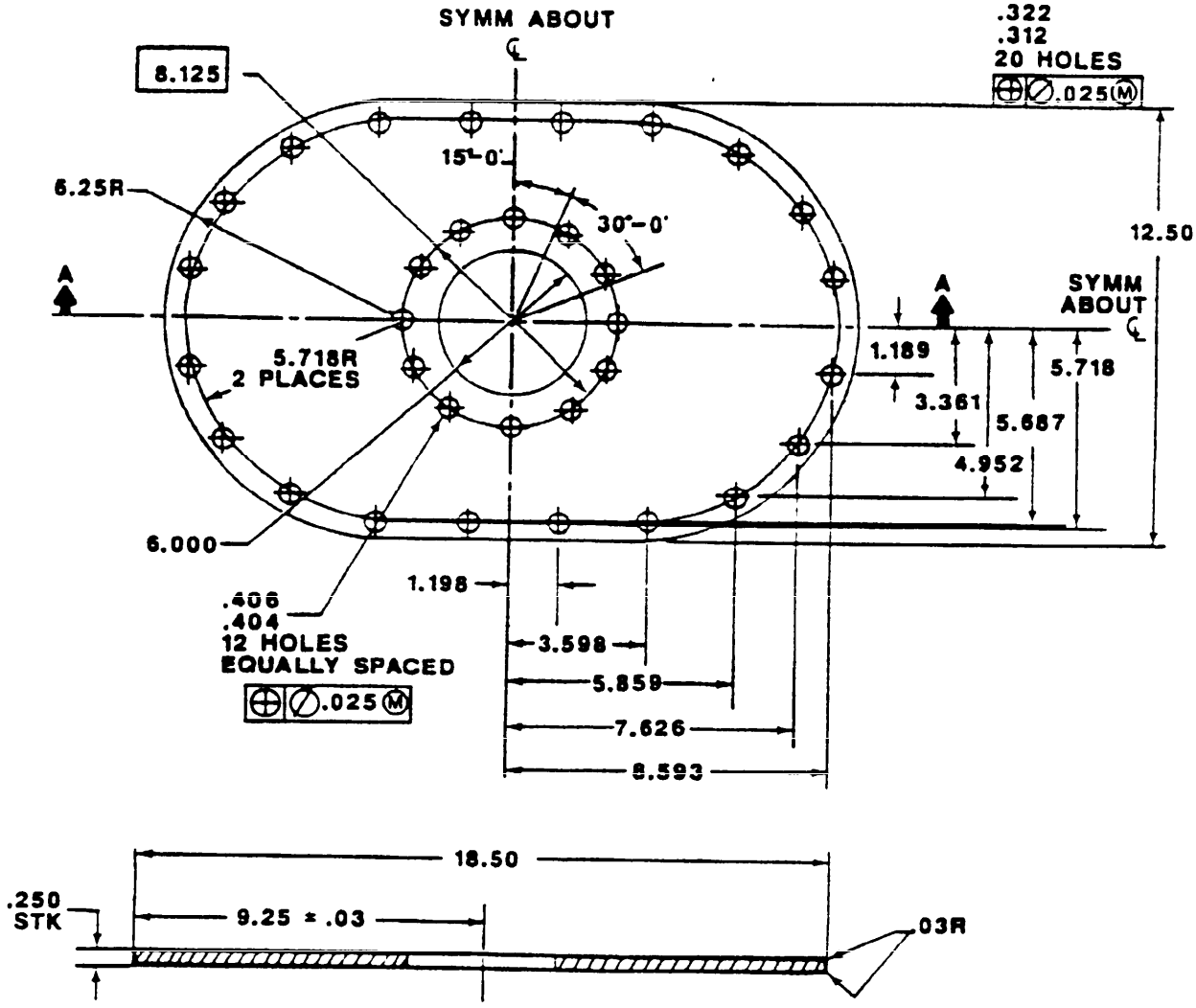
NOTES:

1. UNLESS OTERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
   TOLERANCES: .XX ± .06
   .XXX ± .015
   ANGLES ± 2°0'
2. MATERIAL: AL ALLOY.
3. ANODIZE AND DYE IN ACCORDANCE WITH MIL-A-8625, TYPE II, OR CHEMICAL CONVERSION COATED IN ACCORDANCE WITH MIL-C-5541, CLASS 1A. COLOR SHALL APPROXIMATE COLOR OF TANK FABRIC.

FIGURE 7. Oval closure plate-compression fitting, 3,000, 10,000 and 20,000 gal. tanks, filler.

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SECTION A-A

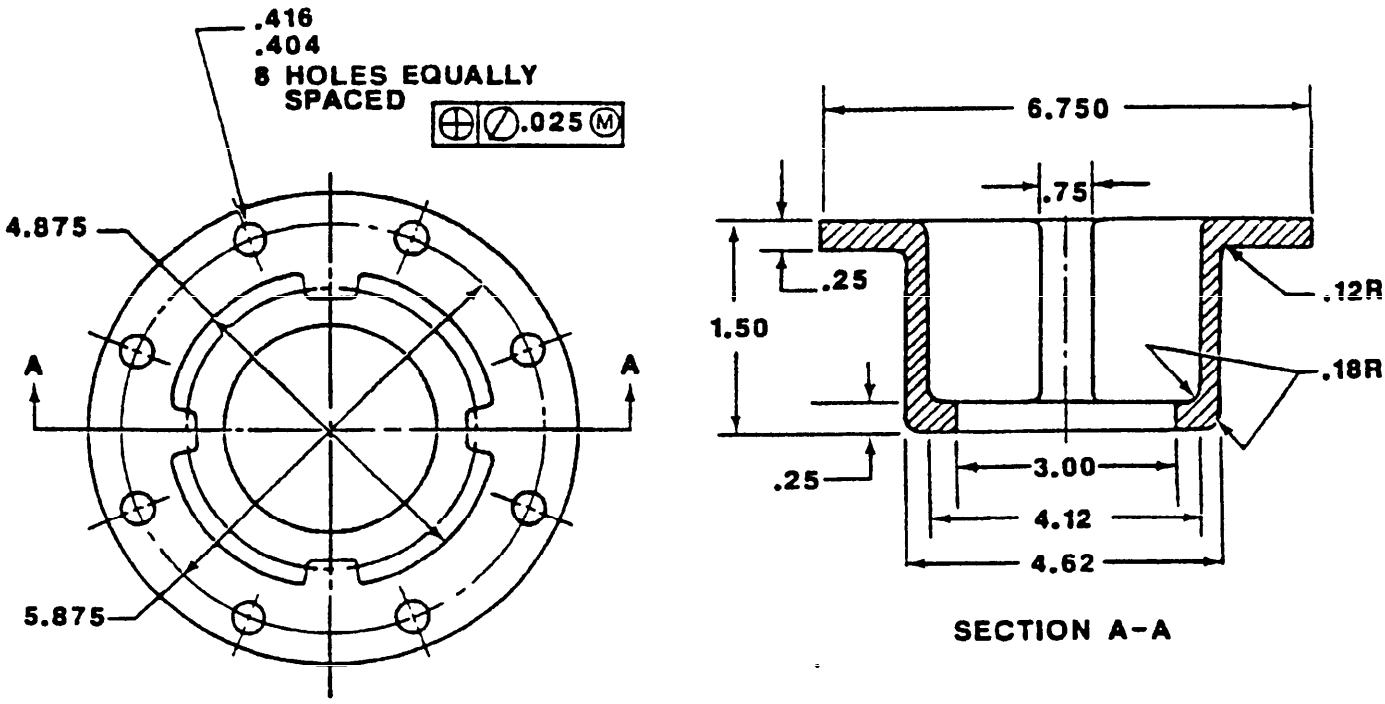
NOTES:

1. UNLESS OTERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
   TOLERANCES: .XX ± .06
   .XXX ± .015
   ANGLES ± 2°0'
2. MATERIAL: AL ALLOY.
3. ANODIZE AND DYE IN ACCORDANCE WITH MIL-A-8625, TYPE II, OR CHEMICAL CONVERSION COATED IN ACCORDANCE WITH MIL-C-5541, CLASS 1A. COLOR SHALL APPROXIMATE COLOR OF TANK FABRIC.

FIGURE 8. Oval closure plate-compression fitting, 3,000, 10,000 and 20,000 gal. tanks, discharge.

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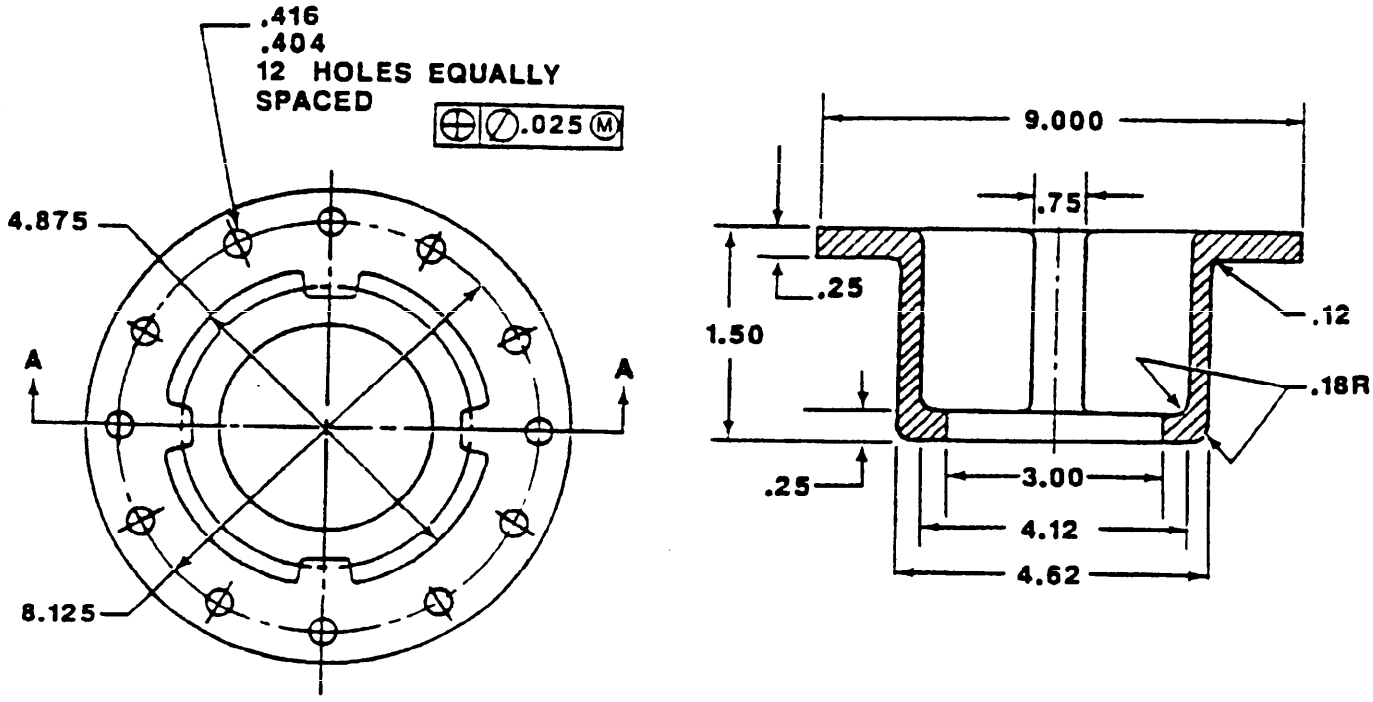


NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. TOLERANCES: .XX = ± .03
   .XXX = ± .015
   DRAFT ANGLES = 3° MAX.
3. MATERIAL: ALUMINUM ALLOY, ANODIZED IN ACCORDANCE WITH MIL-A-8625, TYPE II, CHEMICAL CONVERSION COATED IN ACCORDANCE WITH MIL-C-5541, CLASS 1A.

FIGURE 9. Suction stub, 4 inch, 3,000, 10,000, and 20,000 gal. tanks, filler.

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NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. TOLERANCES: .XX = ± .03
   .XXX = ± .015
   DRAFT ANGLES = 3° MAX.
3. MATERIAL: ALUMINUM ALLOY, ANODIZED IN ACCORDANCE WITH MIL-A-8625, TYPE II, CHEMICAL CONVERSION COATED IN ACCORDANCE WITH MIL-C-5541, CLASS 1A.

FIGURE 10. Suction stub, 6 inch, 3,000, 10,000, and 20,000 gal. tanks, filler.

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NOTES

1. LUBRICATE O-RING WITH SILICONE COMPOUND PER MIL-S-8660 BEFORE ASSEMBLY.

FIND NO.	PART NO. OR IDENT NO.	QTY	DESCRIPTION
1	FIGURE 12	1	VENT AND DRAIN FITTING COMPRESSION TYPE
2	M25988/1-250	1	"O" RING GASKET MIL-R-25988/1, MIL-R-25988
3	MS27183-10	8	WASHER,PLAIN,1/4 INCH STD
4	MS90728-9	8	SCREW,CAP HEXAGON HEAD .25-20 UNC,1.125 LONG MINIMUM
5	MS27023-21	1	COUPLING HALF,QUICK DISCONNECT CAMLOCKING TYPE,MALE FLANGED
6	MS27024-11	1	COUPLING HALF,QUICK DISCONNECT CAMLOCKING TYPE,FEMALE THREADED
7	MS27028-11	1	DUST CAP,2" WITH 12 INCH LG SECURITY CHAIN STYLE B
8		1	PIPE,2 INCH NOM. SCHEDULE 40
9		1	FLAME ARRESTOR
10	49234/EX13338	1	RELIEF CAP WITH GASKET
11	13228E1768-6	2	GASKET-COUPLING HALF,2" SIZE

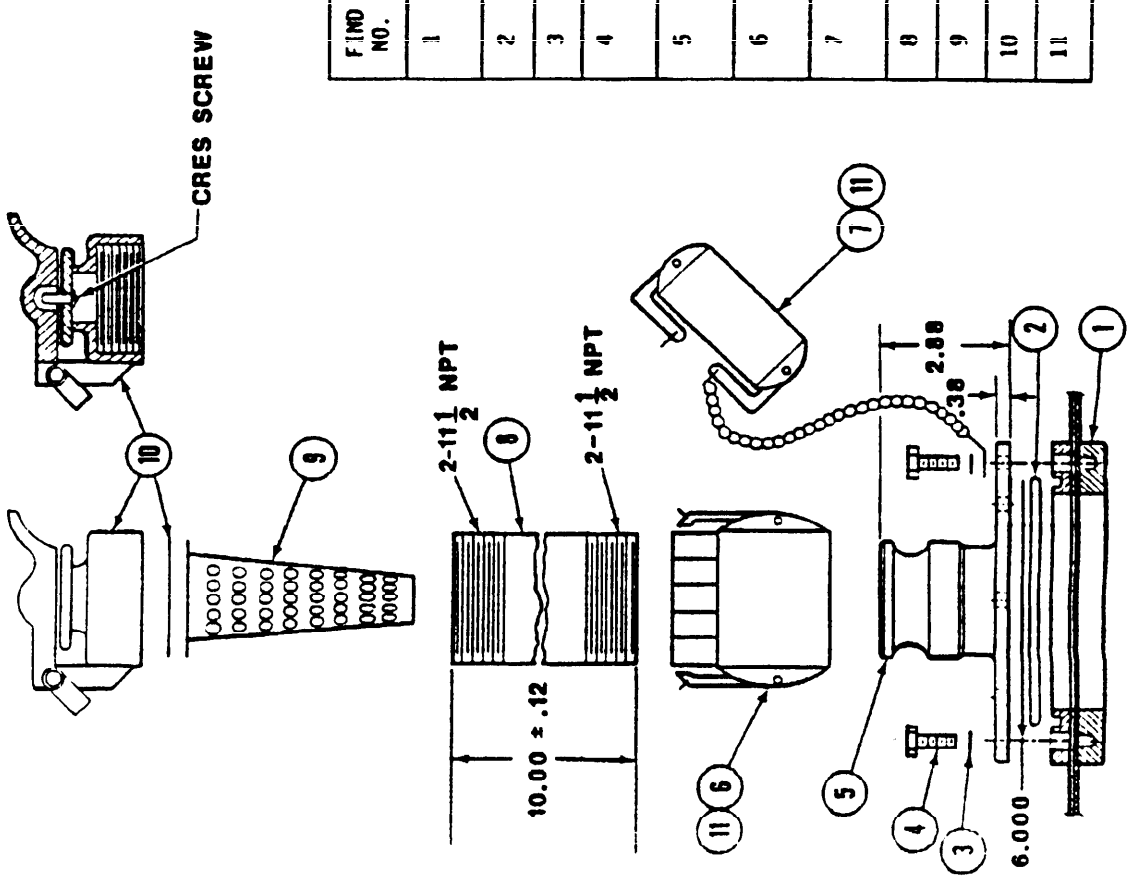
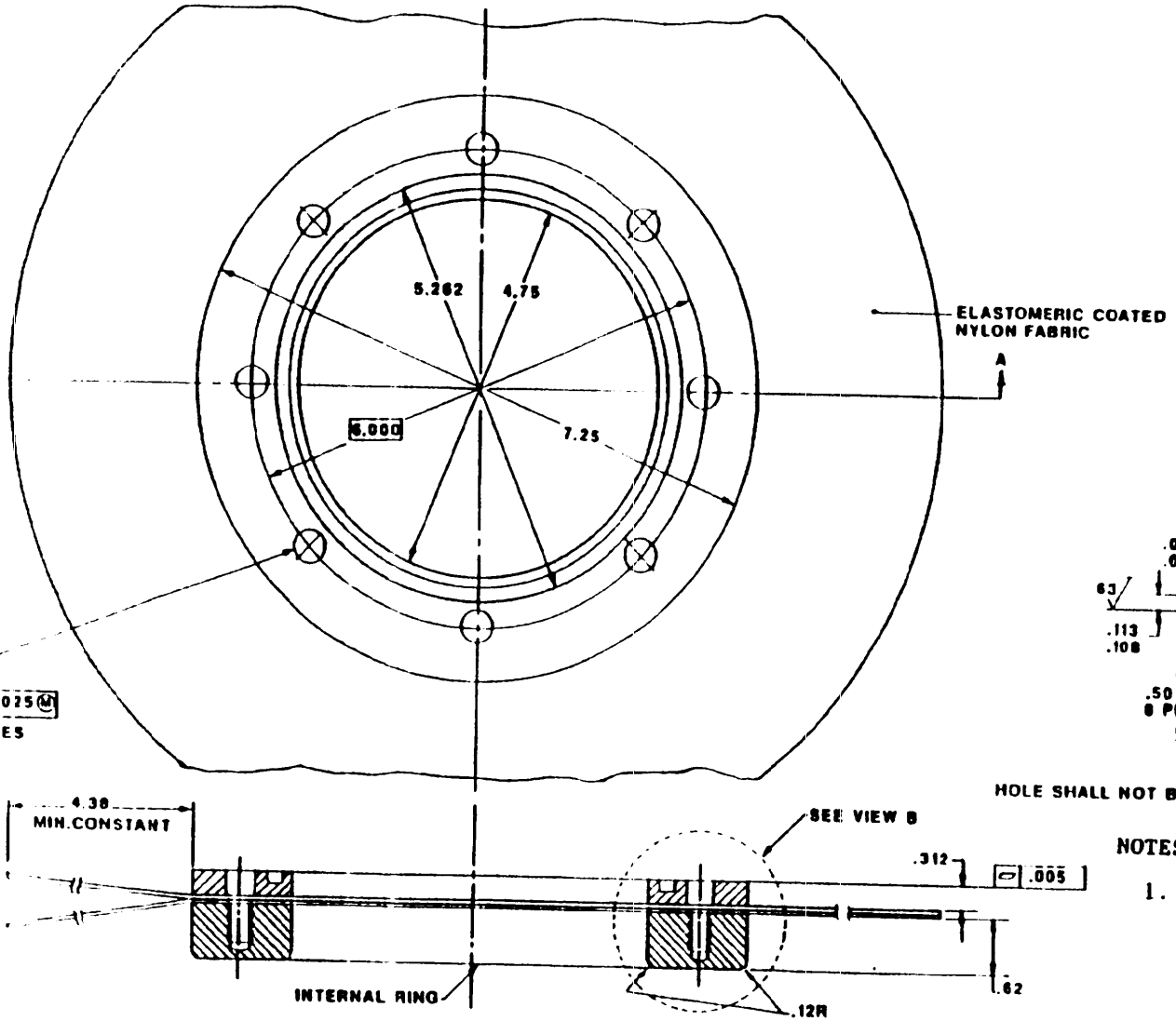


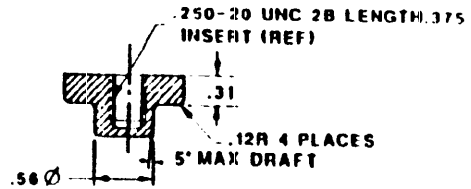
FIGURE 11. Vent fitting assembly.

X-4765A

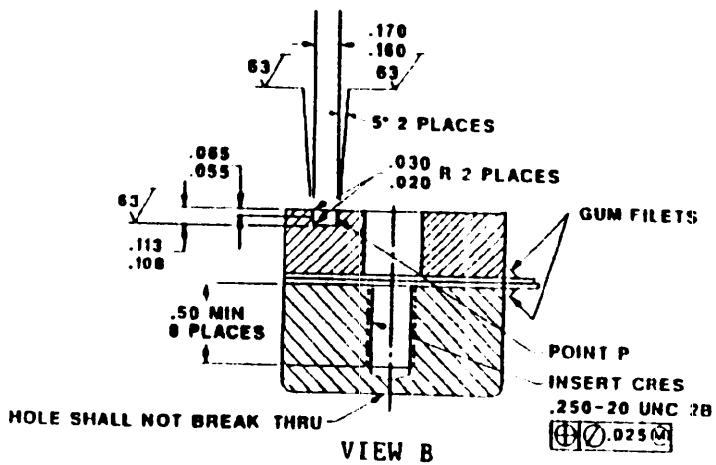
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SECTION A-A



OPTIONAL CAST BOSS DESIGN
(8 PLACES)



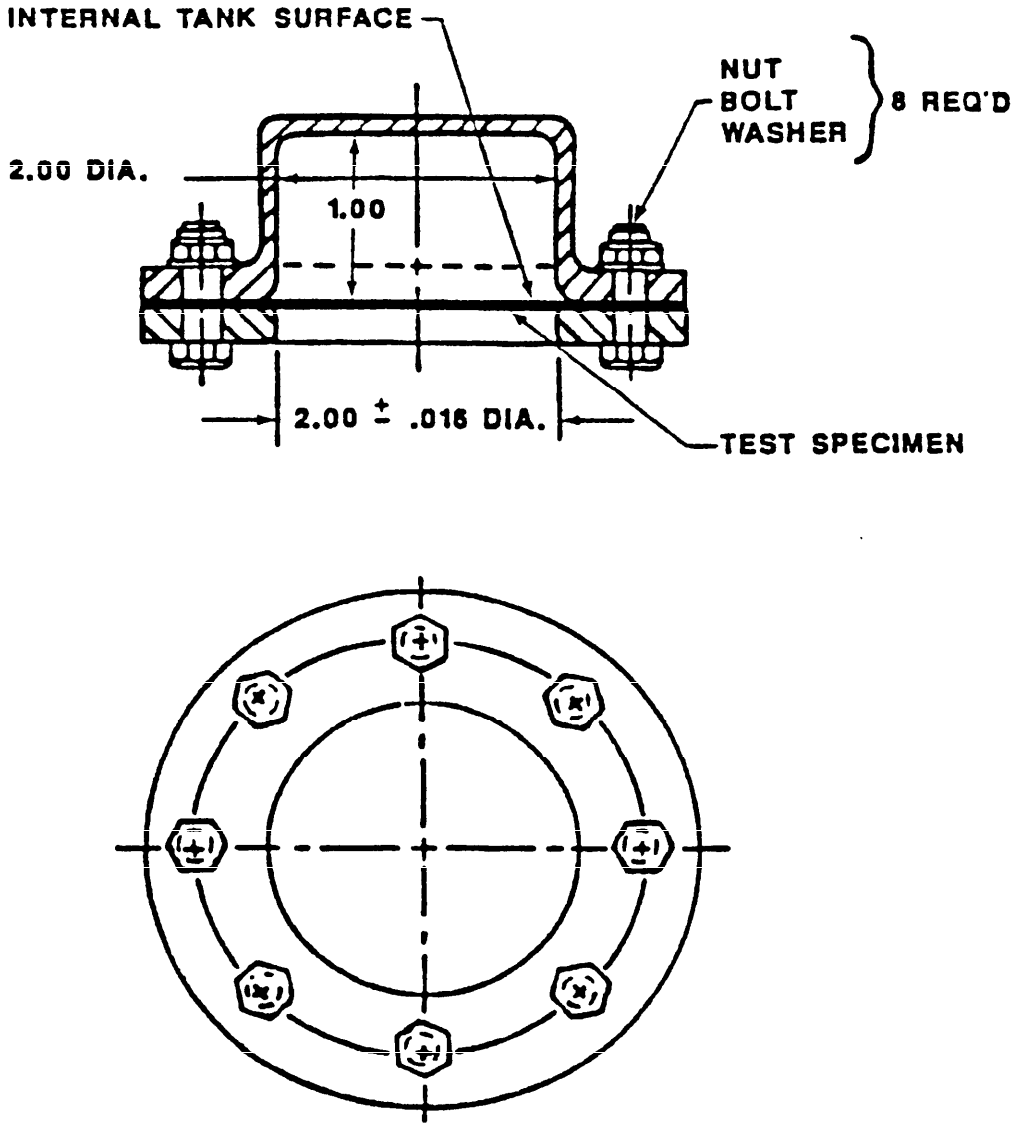
VIEW B

NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
   TOLERANCES: .XX = ± 06
   .XXX = ± 015
   ANGLES = ± 2°
2. ALUMINUM PARTS SHALL BE ANODIZED IN ACCORDANCE WITH MIL-A-8625, TYPE II.
3. ADHESIVE BOND FABRIC AND RINGS.

FIGURE 12. Vent and drain fi[illegible]ng-compression type.

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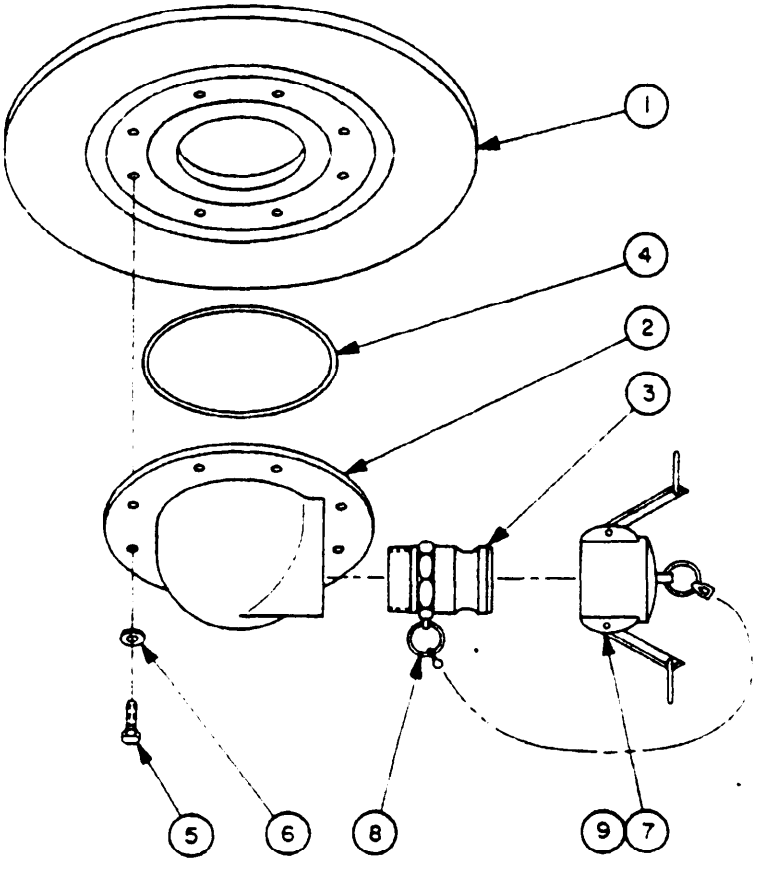
1. UNLESS OTERWISE SPECIFIED DIMENSIONS ARE IN INCHES.

   TOLERANCES: .XX ± .06

2. MATERIAL SHALL BE ALUMINUM ALLOY.

FIGURE 13. Diffusion cup assembly.

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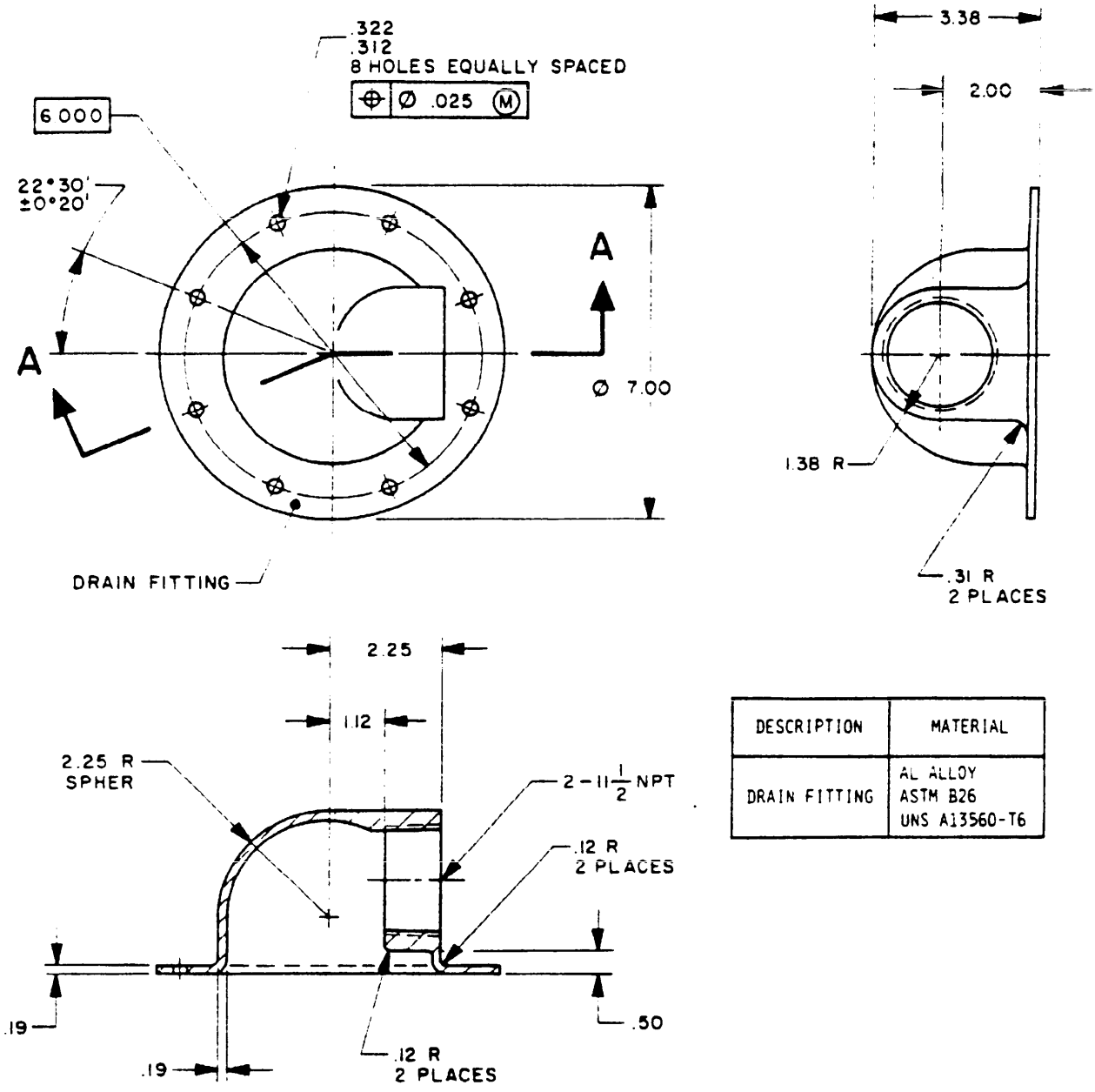
FIND NO.	PART NO. OR IDENT NO.	QTY	DESCRIPTION
1	FIGURE	1	VENT AND DRAIN FITTING, COMPRESSION TYPE
2	FIGURE	1	DRAIN FITTING
3	MS27022-11	1	COUPLING HALF, MALE
4	M25988/1-250	1	"O" RING, MIL-R-25988/1, MIL-R-25988
5	MS90728-8	8	SCREW, CAP, HEXAGON HEAD, 1/4-20 UNC, 1.00 INCH LONG
6	MS27183-10	8	WASHER, FLAT, .281 ID
7	MS27028-11	1	COUPLING HALF, CAP
8	13228E1762-5	1	RING, SPLIT
9	13228E1768-6	1	GASKET, COUPLING HALF

NOTES:

1. CADMIUM PLATING SHALL BE IN ACCORDANCE WITH QQ-P-416, CLASS 1, TYPE II.
2. LUBRICATE O-RING WITH SILICONE COMPOUND IN ACCORDANCE WITH MIL-S-8660 BEFORE ASSEMBLING HARDWARE.

FIGURE 14. Drain fitting assembly.

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DESCRIPTION	MATERIAL
DRAIN FITTING	AL ALLOY ASTM B26 UNS A13560-T6

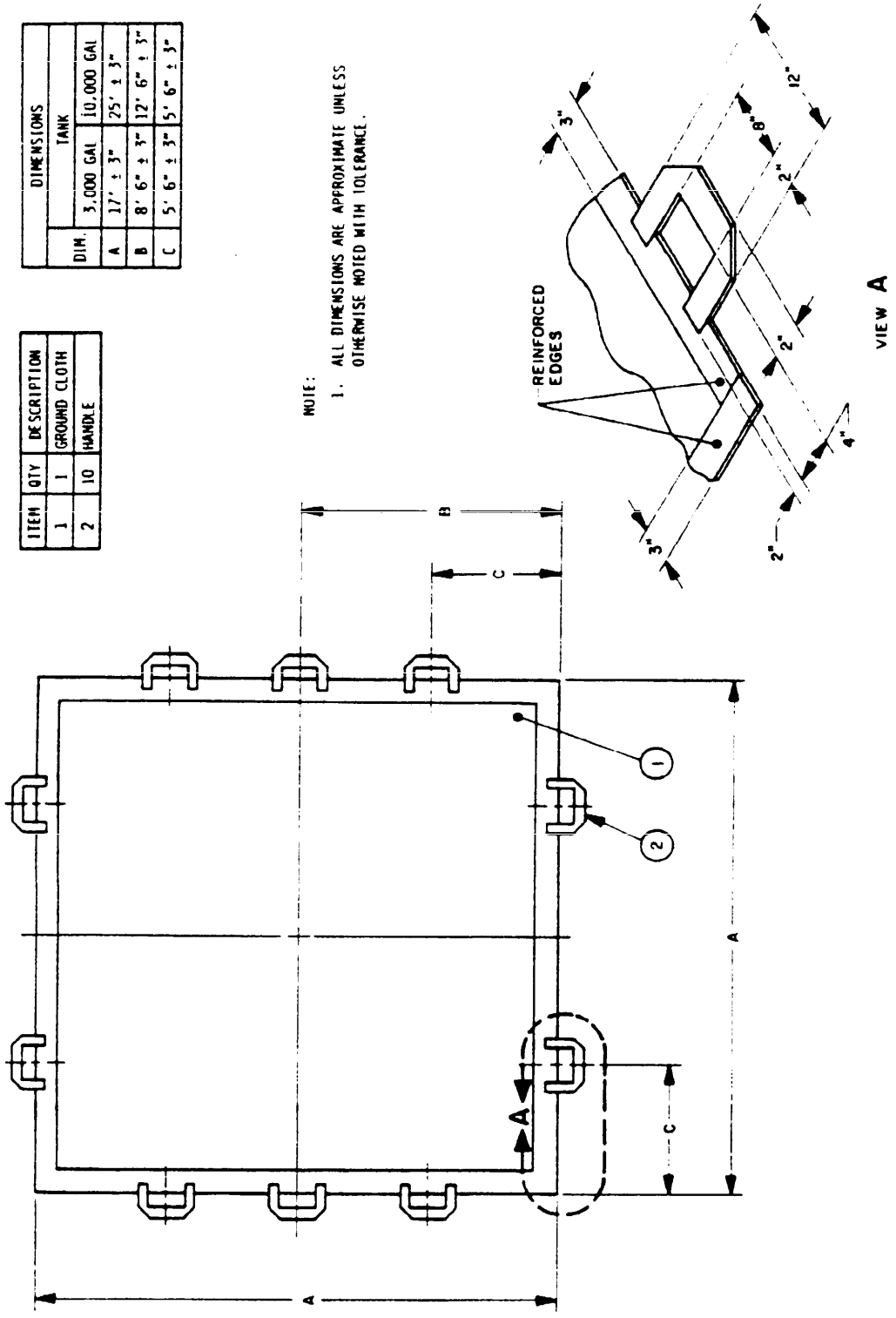
NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
   TOLERANCES XX ± .06
   DRAFT ANGLES = 3° MAX
2. ALUMINUM ALLOY PART SHALL BE ANODIZED IN ACCORDANCE WITH MIL-A-8625, TYPE II.

FIGURE 15. Drain fitting.

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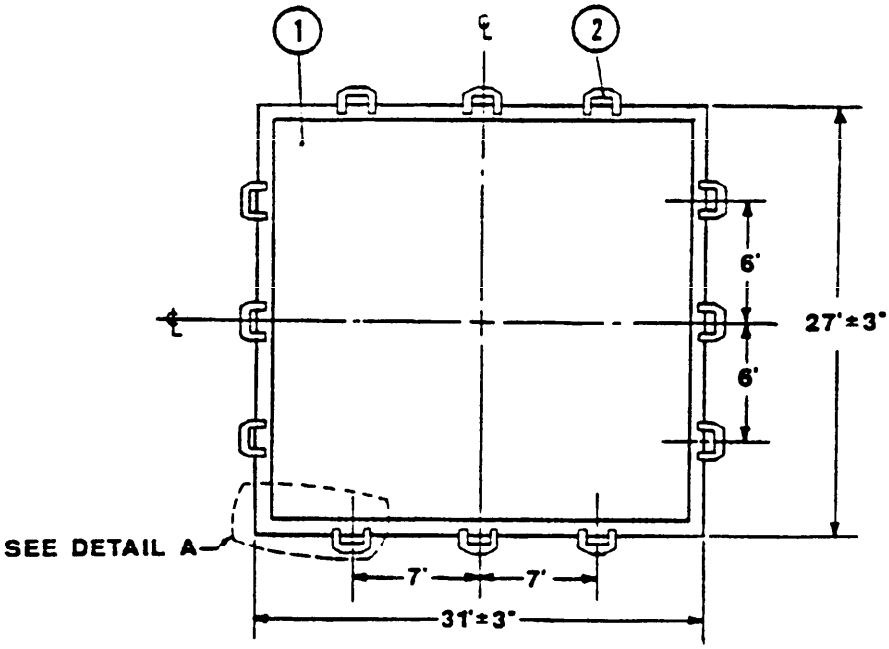
X-4769A



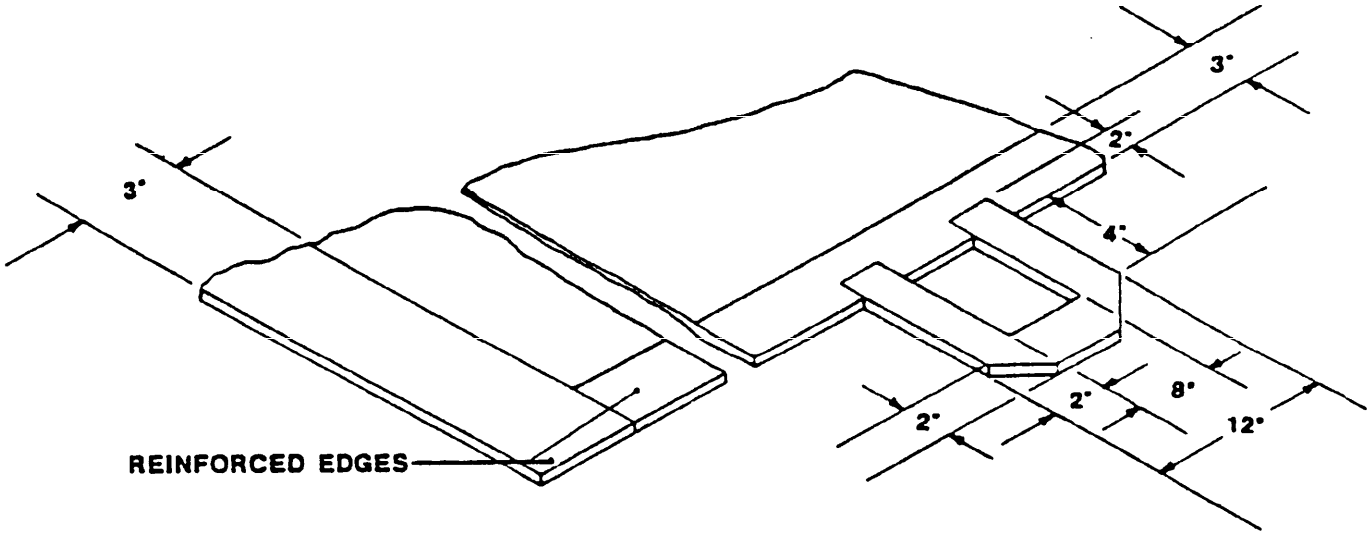
X-4848

FIGURE 16. Ground cloth, 3,000 and 10,000 gal. tanks.

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ITEM	QTY	DESCRIPTION
1	1	GOUND CLOTH
2	12	HANDLE



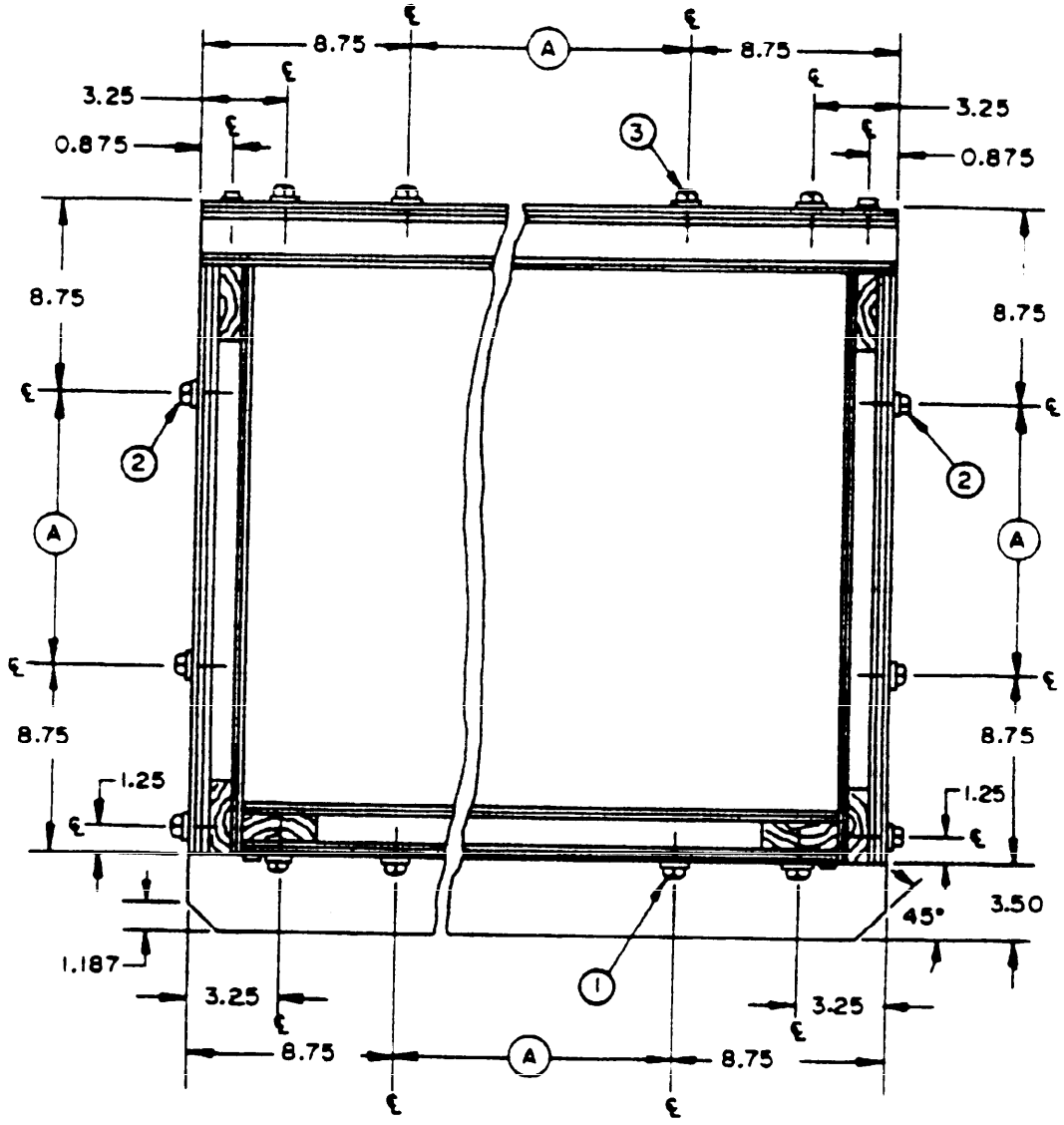
DETAIL A

NOTE

1. ALL DIMENSIONS ARE APPROXIMATE UNLESS OTHERWISE NOTED WITH TOLERANCE

FIGURE 17. Ground cloth, 20,000 gal. tank.

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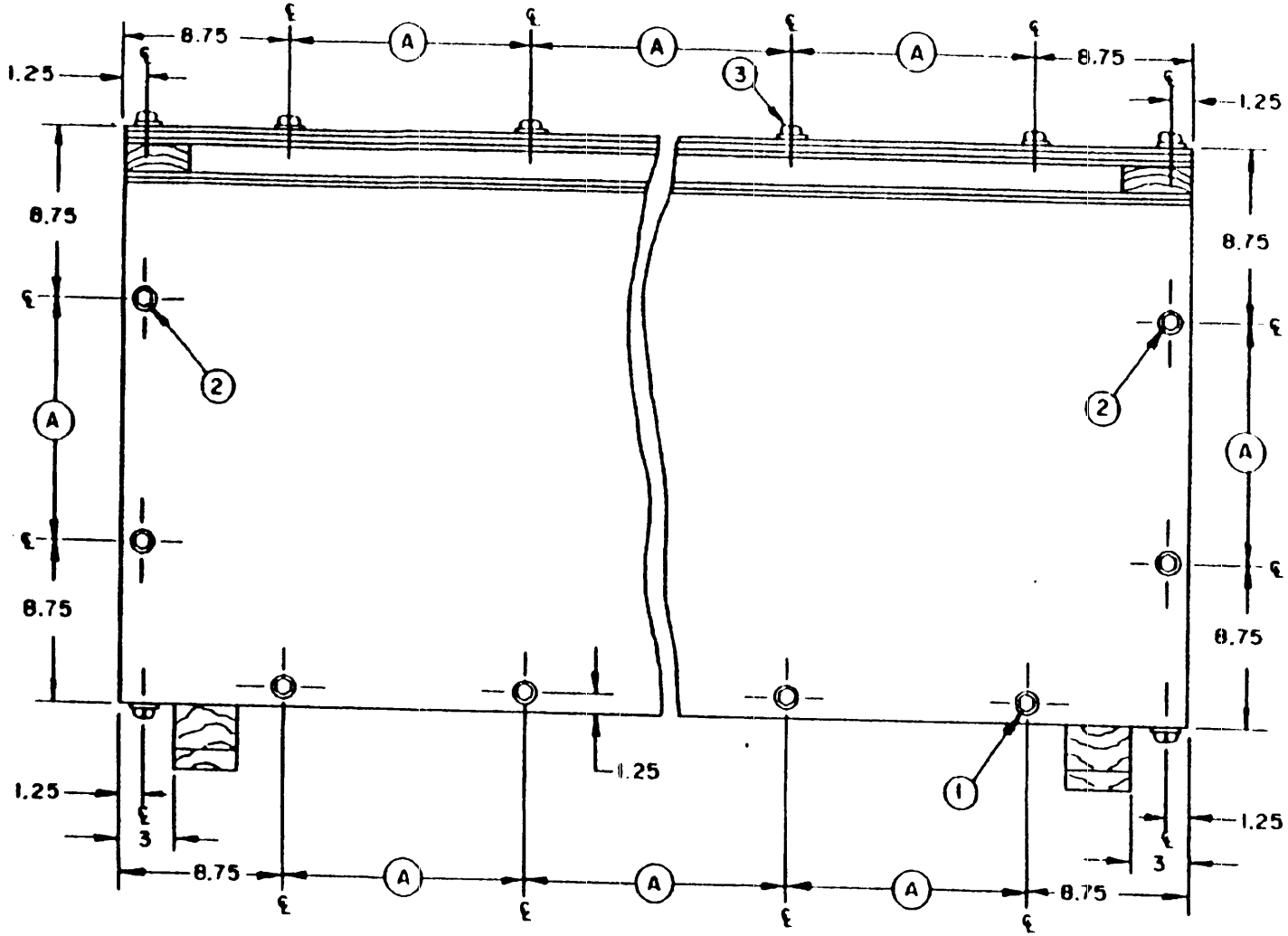
NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) BOTTOM ASSEMBLY TO END PANEL.
   - (2) SIDE PANEL TO END PANEL.
   - (3) TOP PANEL TO END PANEL.

FIGURE 18. End View of Box.

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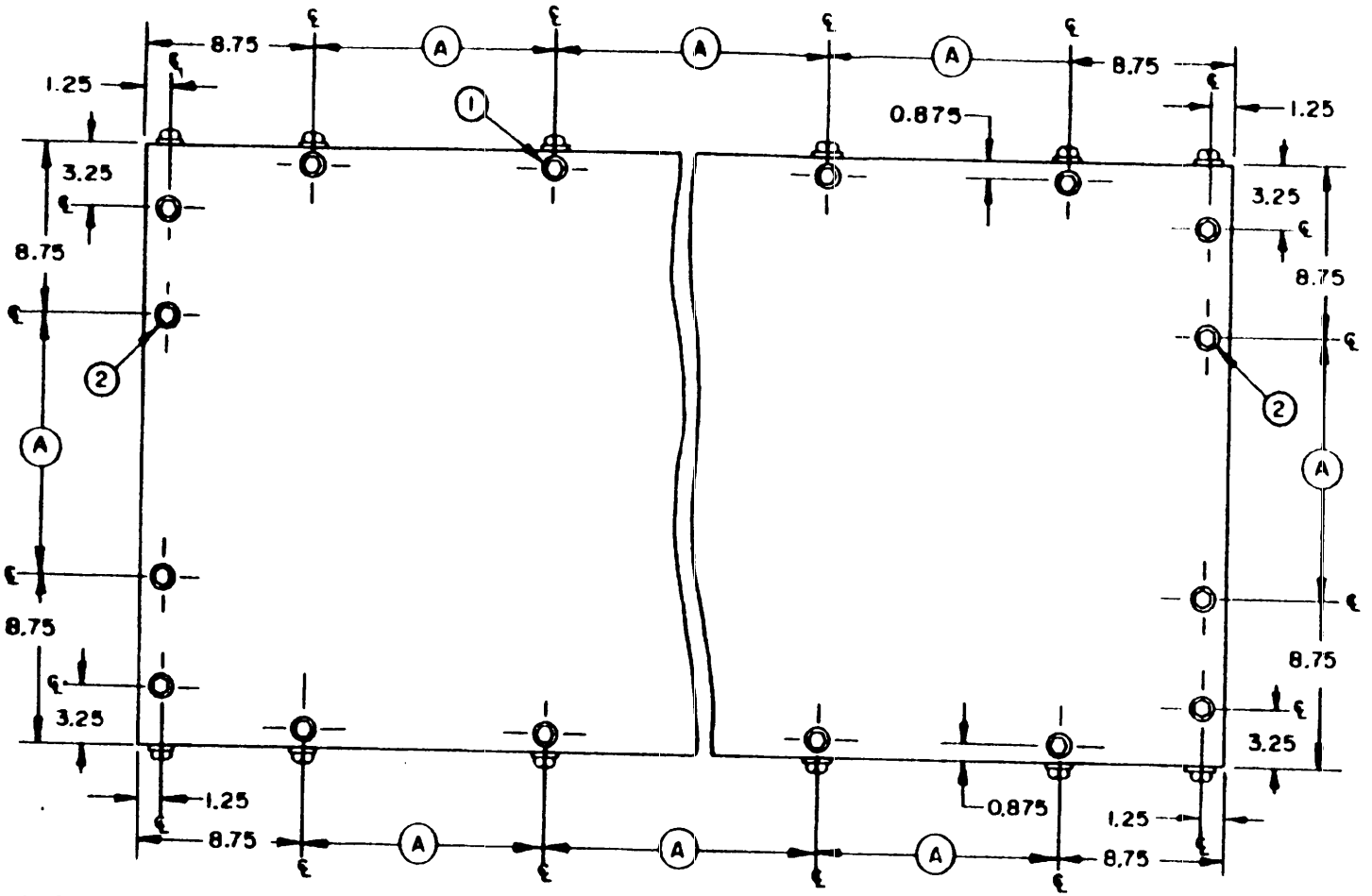


NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) SIDE PANEL TO BOTTOM PANEL.
   - (2) SIDE PANEL TO END PANEL.
   - (3) TOP PANEL TO SIDE PANEL.

FIGURE 19. Side View of Box.

X-4708B



NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:

   (1) TOP PANEL TO SIDE PANEL.

   (2) TOP PANEL TO END PANEL.

FIGURE 20. Top View of Box.

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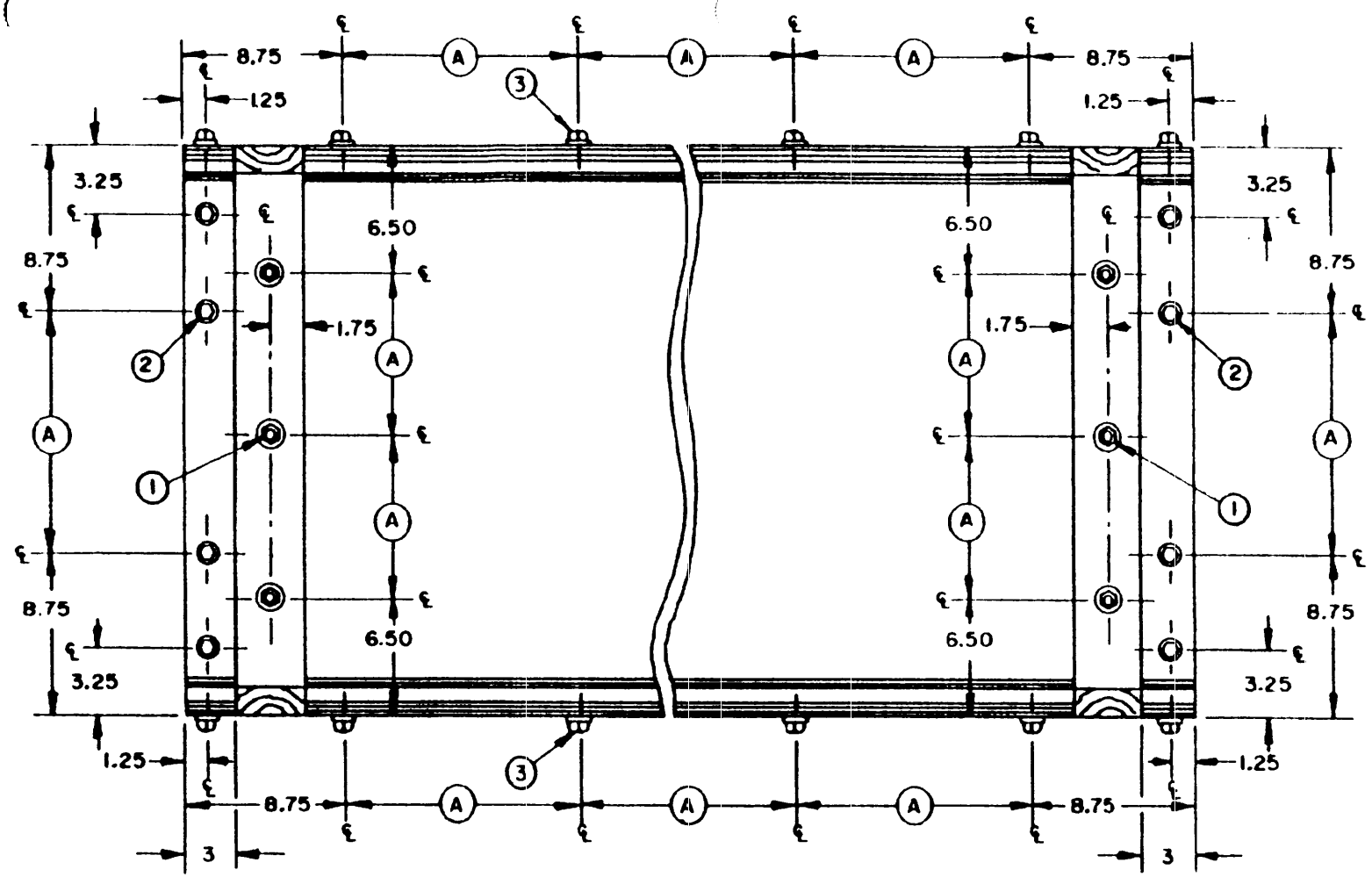
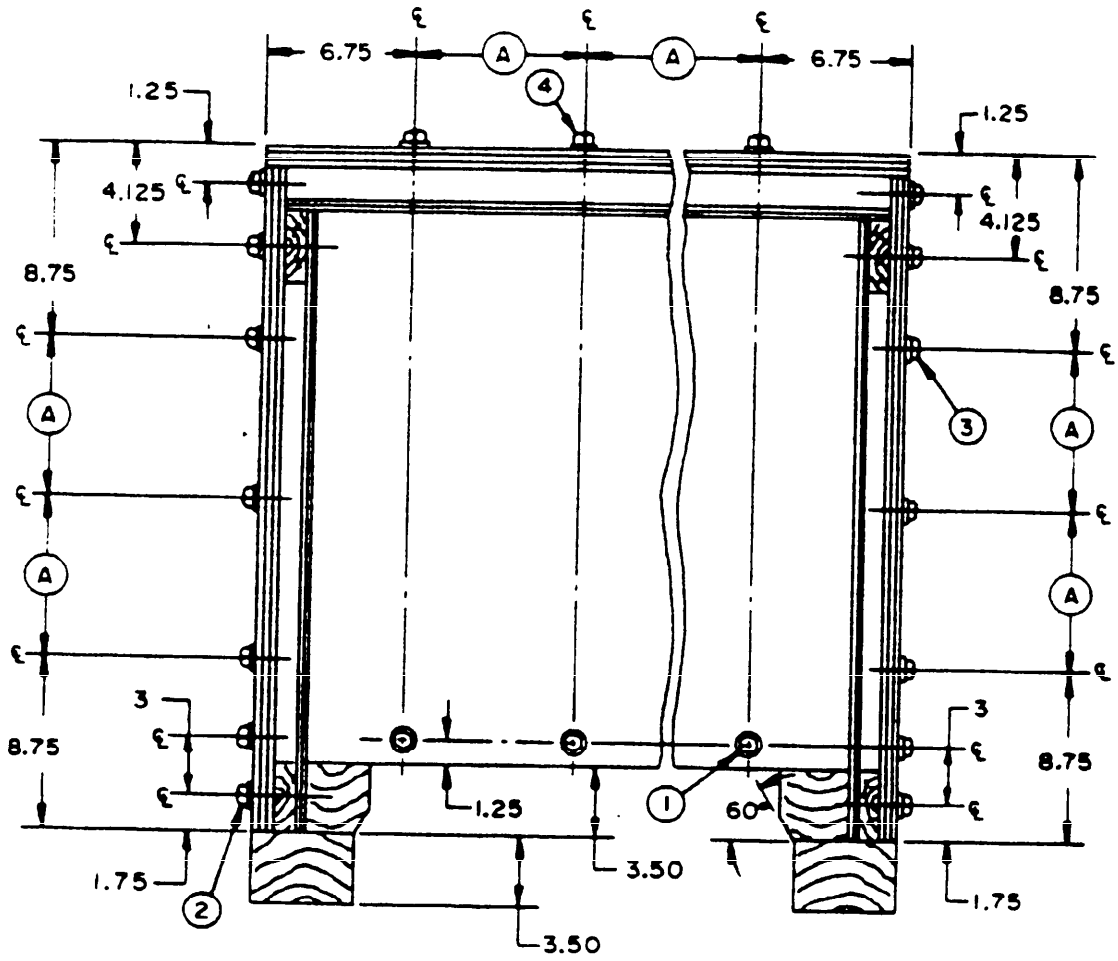


FIGURE 21. Bottom View of Box.

X-4710A

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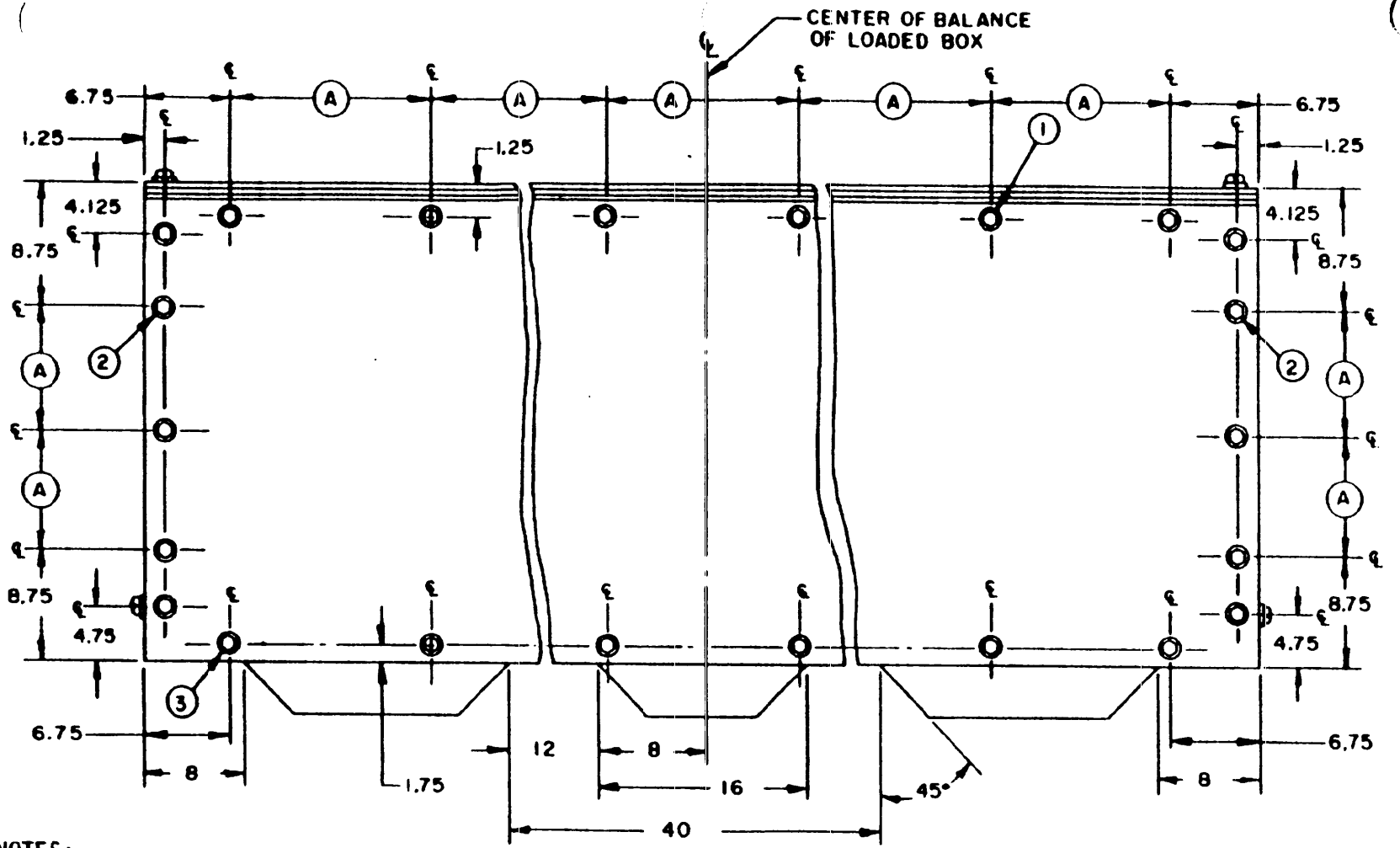
NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) END PANEL TO BASE.
   - (2) SIDE PANEL TO BASE.
   - (3) SIDE PANEL TO END PANEL.
   - (4) TOP PANEL TO END PANEL.

FIGURE 22. End View of Box With Outside Length Greater Than 96 Inches.

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X-4711



NOTES:

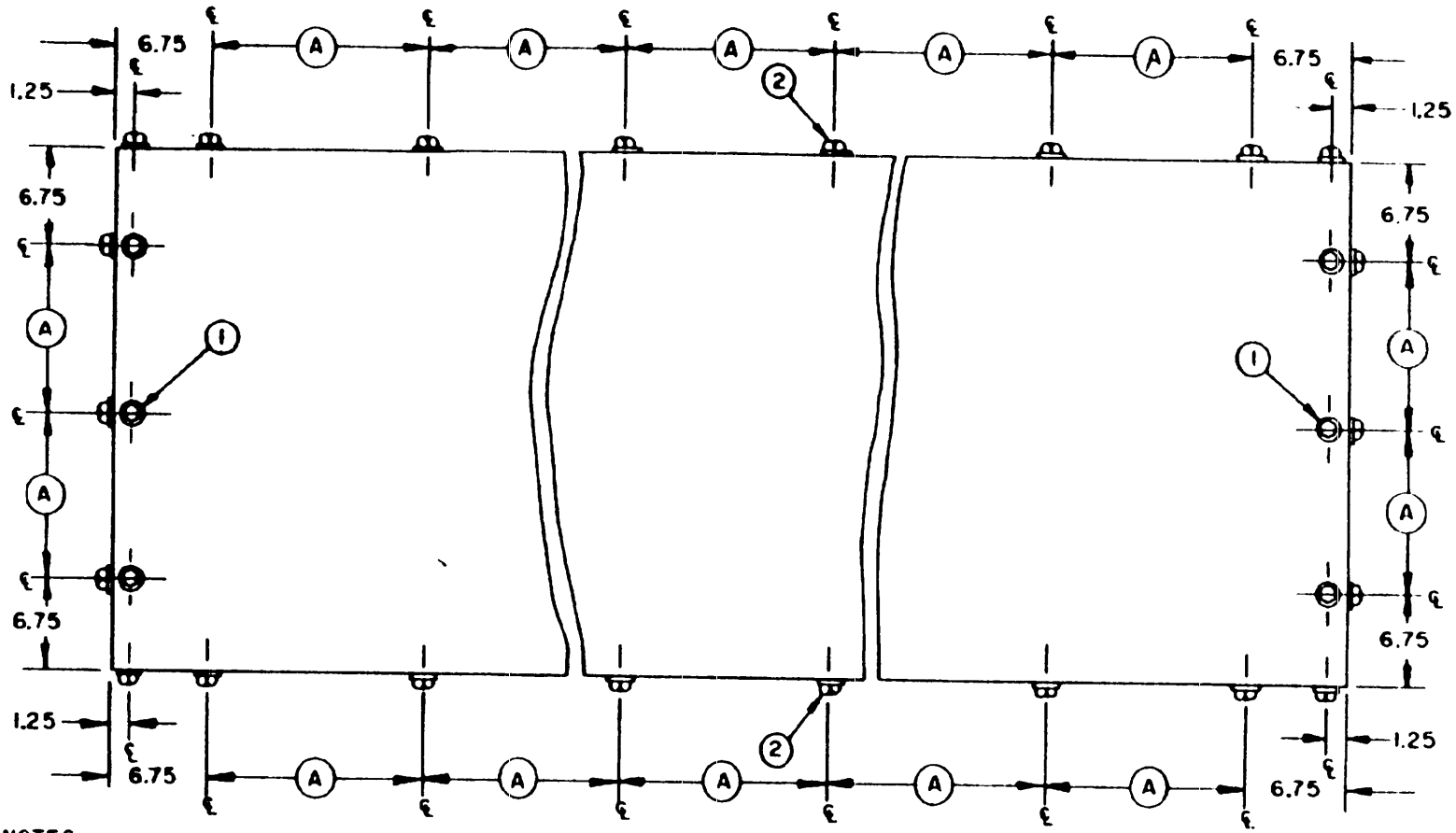
1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:

   (1) SIDE PANEL TO TOP PANEL.
   (2) SIDE PANEL TO END PANEL.
   (3) SIDE PANEL TO BASE.

FIGURE 23. Side View of Box With Outside Length Dimension Greater Than 96 Inches.

X-4712

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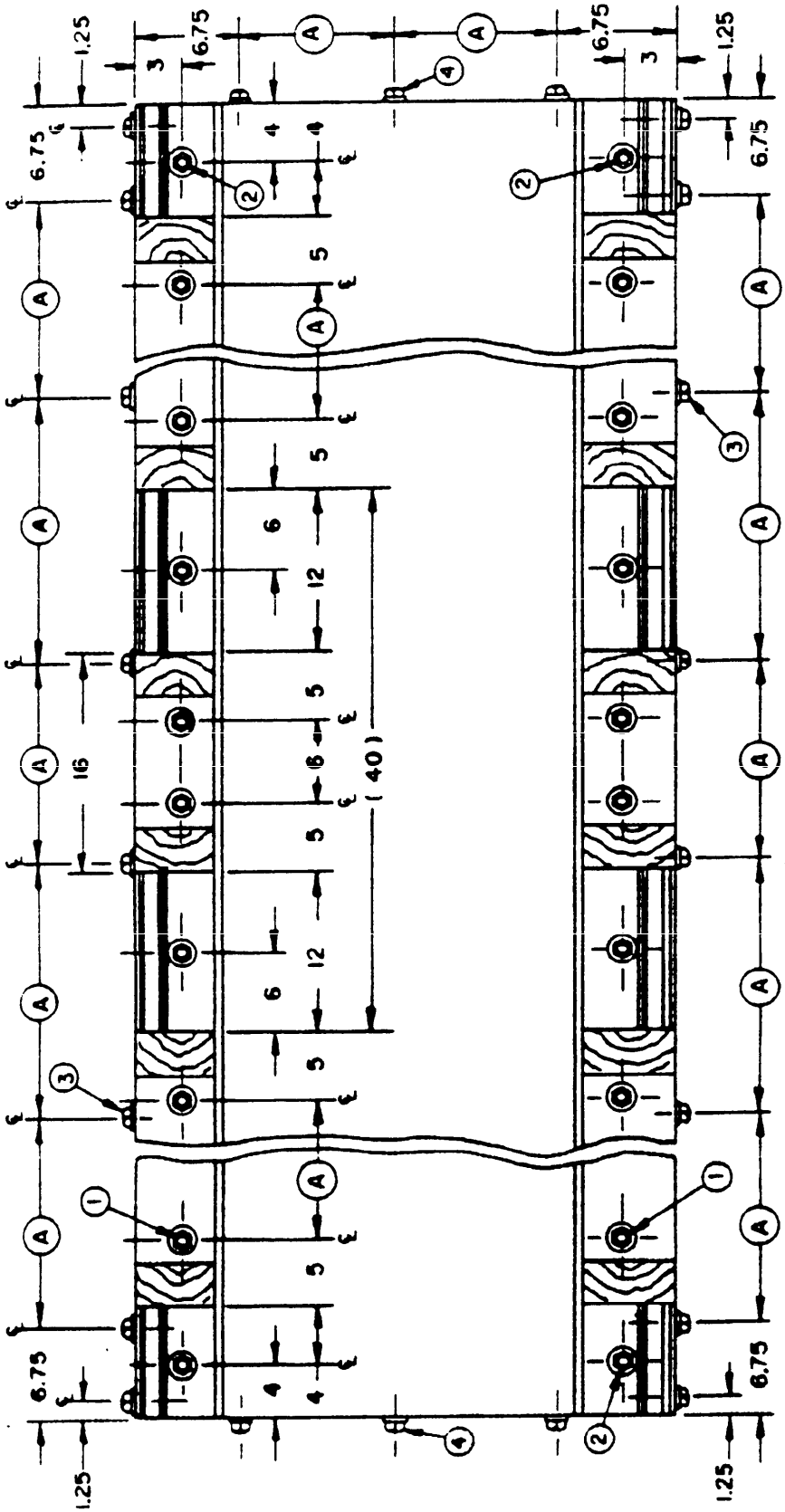


NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) TOP PANEL TO END PANEL.
   - (2) SIDE PANEL TO TOP PANEL.

FIGURE 24. Top View of Box With Outside Length Dimension Greater Than 96 Inches.

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS AND CARRIAGE BOLTS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) RUBBING STRIPS AND SKIDS TO BASE.
   - (2) HEADER TO BASE.
   - (3) SIDE PANEL TO BASE.
   - (4) END PANEL TO BASE.

FIGURE 25. Bottom View of Box With Outside Length Dimension Greater Than 96 Inches.

X-4714

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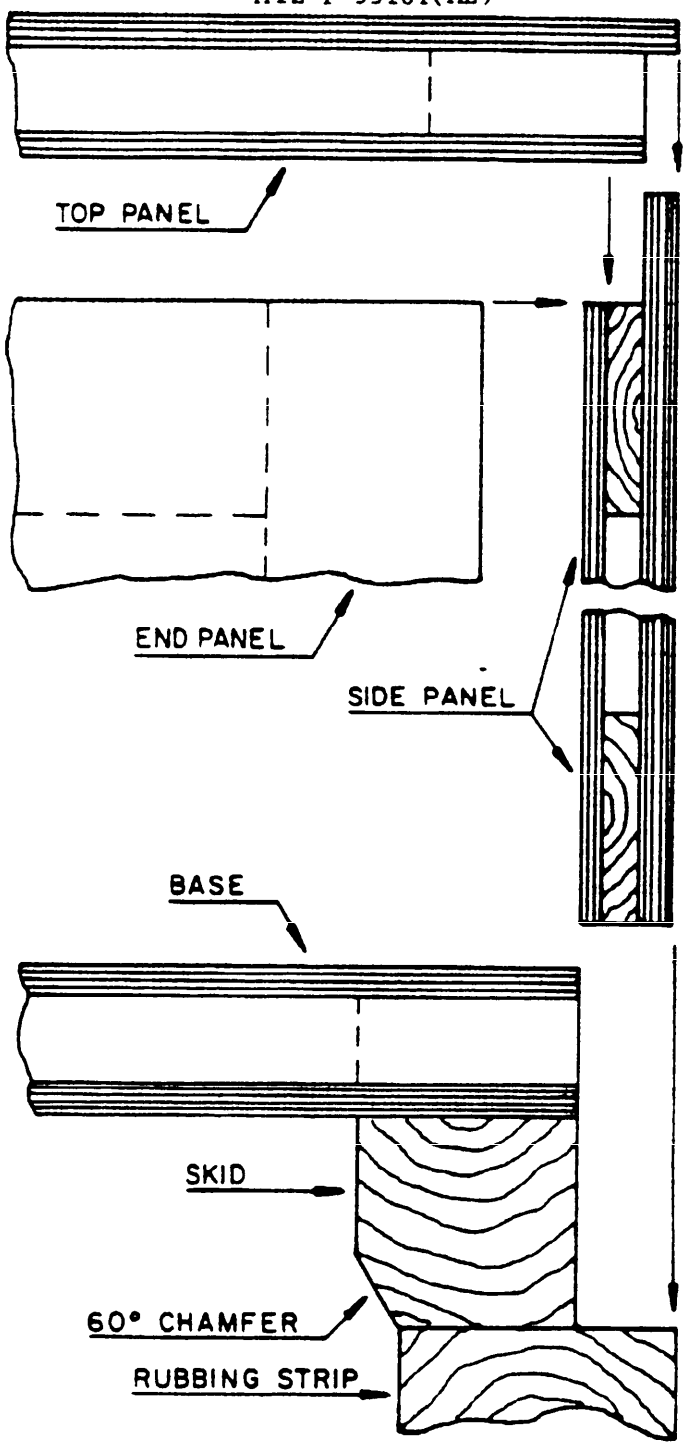
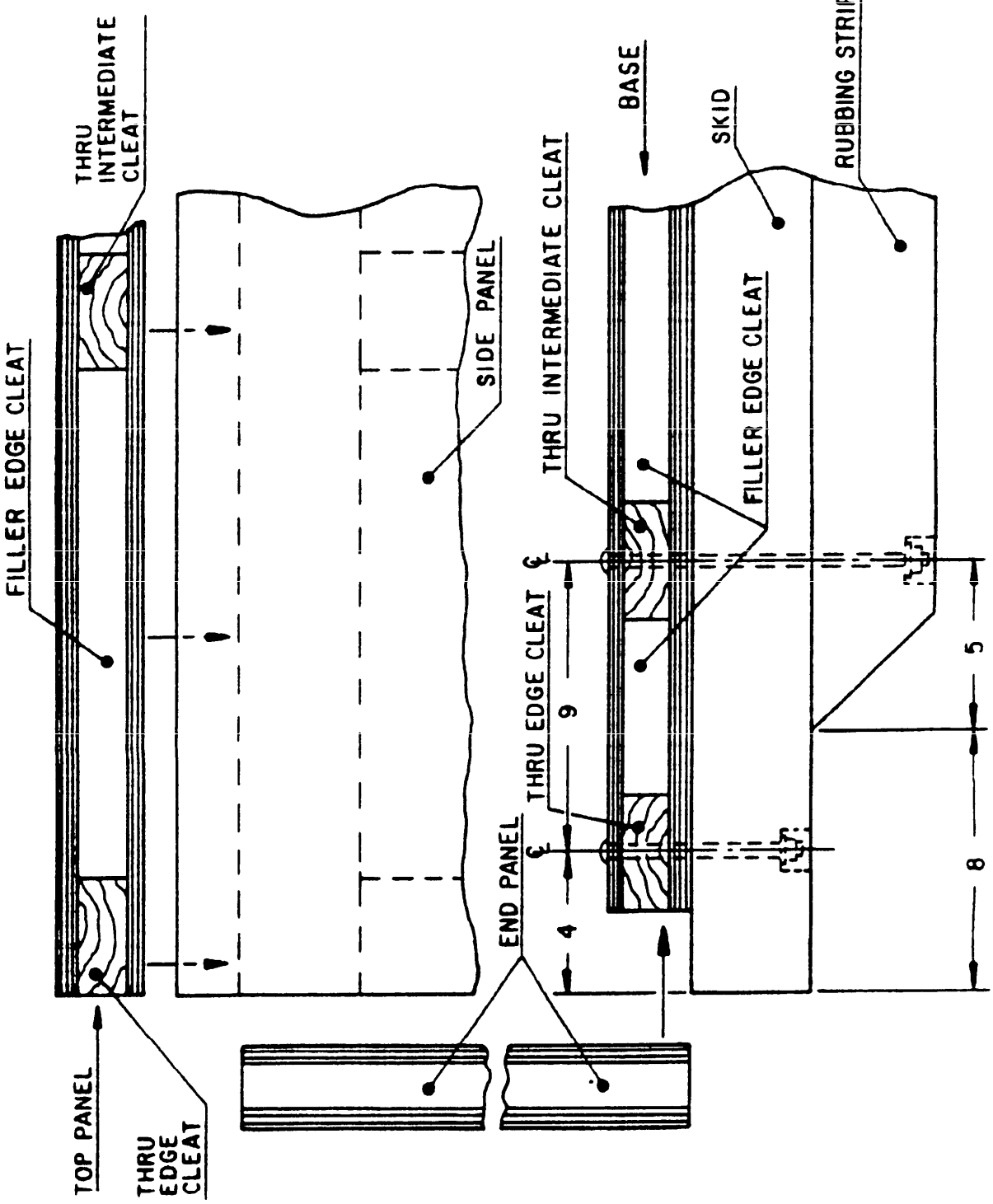


FIGURE 26. Assembly Method (End View) for Box Over 96 Inches in Length.

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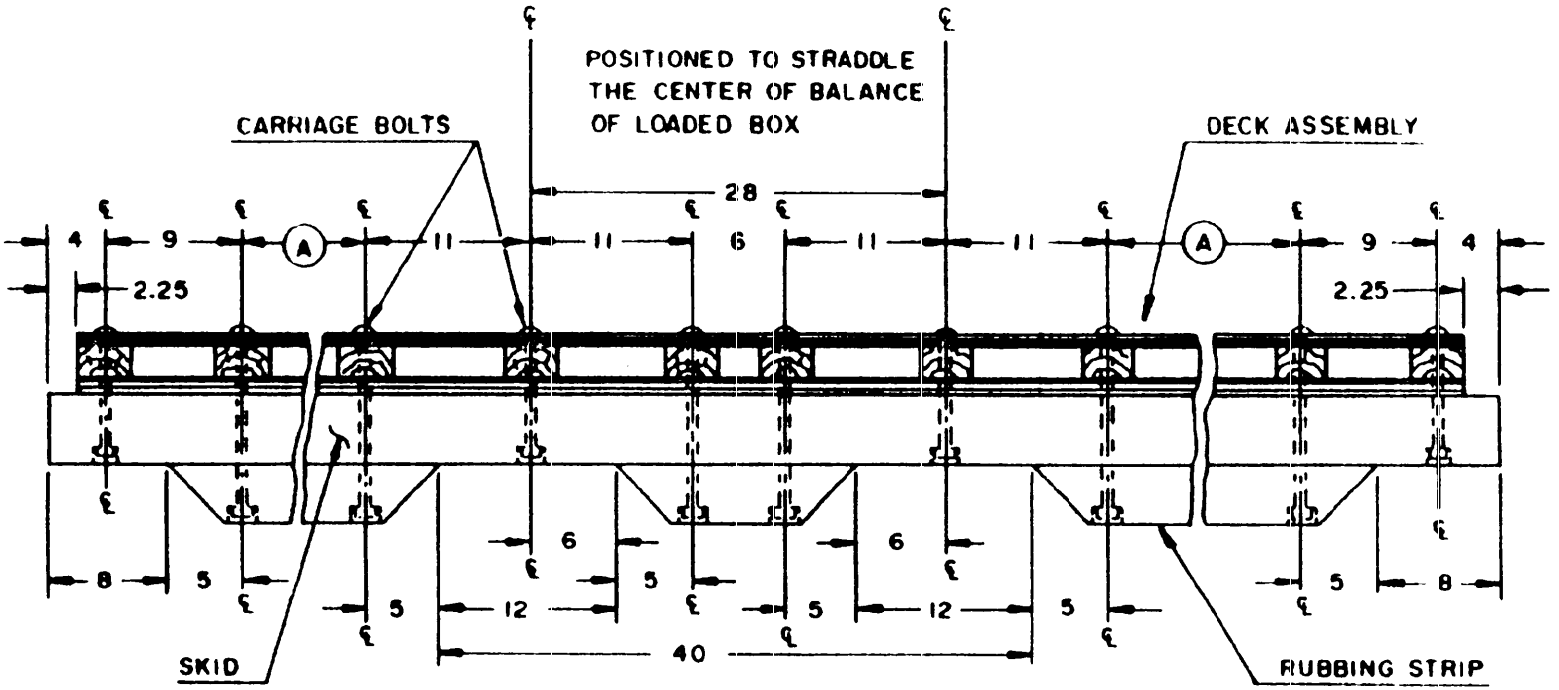
X-4716

FIGURE 27. Assembly Method (Side View) for Box Over 96 Inches in Length.

NOTE:

1. ALL DIMENSIONS ARE IN INCHES.

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSION (A) BETWEEN THRU INTERMEDIATE CLEATS AND INTERMEDIATE CARRIAGE BOLTS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.

FIGURE 28. Fabrication Method for Base (Side View) for Box Over 96 Inches in Length.

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Consumer Information Sheet

INORGANIC ARSENICAL PRESSURE-TREATED WOOD

(Including: CCA; ACA, and ACZA)

CONSUMER INFORMATION

This wood has been preserved by pressure-treatment with an EPA-registered pesticide containing inorganic arsenic to protect it from insect attack and decay. Wood treated with inorganic arsenic should be used only where such protection is important.

Inorganic arsenic penetrates deeply into and remains in the pressure-treated wood for a long time. Exposure to inorganic arsenic may present certain hazards. Therefore, the following precautions should be taken both when handling the treated wood and in determining where to use or dispose of the treated wood.

USE SITE PRECAUTIONS

Wood pressure-treated with waterborne arsenical preservatives may be used inside residences as long as all sawdust and construction debris are cleaned up and disposed of after construction.

Do not use treated wood under circumstances where the preservative may become a component of food or animal feed. Examples of such sites would be structures or containers for storing silage or food.

Do not use treated wood for cutting-boards or counter-tops.

Only treated wood that is visibly clean and free of surface residue should be used for patios, decks and walkways.

Do not use treated wood for construction of those portions of beehives which may come into contact with the honey.

Treated wood should not be used where it may come into direct or indirect contact with public drinking water, except for uses involving incidental contact such as docks and bridges.

HANDLING PRECAUTIONS

Dispose of treated wood by ordinary trash collection or burial. Treated wood should not be burned in open fires or in stoves, fireplaces, or residential boilers because toxic chemicals may be produced as part of the smoke and ashes. Treated wood from commercial or industrial use (e.g., construction sites) may be burned only in commercial or industrial incinerators or boilers in accordance with state and Federal regulations.

Avoid frequent or prolonged inhalation of sawdust from treated wood. When sawing and machining treated wood, wear a dust mask. Whenever possible, these operations should be performed outdoors to avoid indoor accumulations of airborne sawdust from treated wood.

When power-sawing and machining, wear goggles to protect eyes from flying particles.

After working with the wood, and before eating, drinking, and use of tobacco products, wash exposed areas thoroughly.

If preservatives or sawdust accumulate on clothes, launder before reuse. Wash work clothes separately from other household clothing.

Approved by the U.S. Environmental Protection Agency 9/85

FIGURE 29. An example of an EPA approved set of instructions for proper handling of CCA.

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